

Did Decreasing Residential Segregation Reduce Racial Wealth Inequality?*

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Abstract

In this paper, I study how the Fair Housing Act (FHAAct), which ended lawful housing discrimination in the US in 1968, affected residential segregation and wealth inequality between Black and White households. I first present three stylized facts, showing that during the 1970s, (i) racial residential segregation started to decline, (ii) the relocation of Black families from Black neighborhoods intensified, and (iii) average housing wealth increased rapidly among Black families. To investigate whether the FHAAct was driving these changes, I estimate a heterogeneous agent life-cycle OLG model with spatial equilibrium and endogenous house prices. With this model, I simulate how the FHAAct affected families' relocation across neighborhoods, neighborhood-level house prices, and the Black and White wealth distribution. I find that the FHAAct could explain (i) the decreasing residential segregation, (ii) the observed relocation patterns, and (iii) partly the increasing housing wealth of Black families. My results suggest that the FHAAct substantially decreased racial wealth inequality at the top while, somewhat unexpectedly, it increased racial wealth inequality between the 2nd and 7th deciles of the wealth distribution.

JEL codes: D15, D31, E21, G51, J15, N32, N92, R23

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1 Introduction

Despite numerous ongoing federal programs aiming to decrease racial wealth inequality, today the Black-White wealth gap is nearly as large as it was during the second half of the 1970s. Based on the 2022 wave of the Survey of Consumer Finances, the median non-Hispanic White household has more than six times as much net wealth as the median Black household, a ratio that has not declined since the mid-1980s. Racial wealth differences decreased rapidly during the Civil Rights Movement, and the passage of the Fair Housing Act (FHA) in 1968 was expected to aid this process by enabling Black families to gain better access to mortgages in more integrated neighborhoods ([Aaronson et al., 2021](#)). The Fair Housing Act promoted racial equity in access to housing wealth by banning racial discrimination in housing sales, rentals, or financing and, through these, ended lawful housing segregation in the United States. As I show later, right after the FHA, racial residential segregation declined and racial wealth convergence accelerated, but during the 1980s, racial wealth differences started to rise again. The halt of racial wealth convergence during the 1980s is puzzling, as many scholars, such as [Shapiro \(2017\)](#) and [Rothstein \(2018\)](#), argue that racial segregation was a key element in the buildup of racial wealth differences, suggesting that declining segregation should go hand in hand with declining racial wealth differences.

In this paper, I study whether the end of housing discrimination reduces racial residential segregation and racial wealth inequality. I model the connection between housing discrimination, segregation, and racial wealth inequality within a life-cycle overlapping-generations model, which allows me to rule out the effect of all economic changes other than the FHA. My simulation results reveal that decreasing housing discrimination led to lower levels of racial residential segregation, and racial wealth inequality decreased among already wealthy families. However, I find that reducing housing discrimination could increase racial wealth inequality in the middle of the wealth distribution.

I first present three stylized facts that show sizable changes in neighborhood sorting dynamics and housing wealth accumulation right after the passage of the Fair Housing Act in 1968. First, the persistent increase of residential segregation between Black and non-Black families stopped during the 1970s, and until 2000, racial segregation declined as much as it increased during the Great Migration between 1910 and 1970 (Cutler et al., 1999; Boustan, 2011). Second, Black neighborhoods experienced a significant and unprecedented outflow of Black families during the 1970s, which continued in later decades at a slower pace (Ellen, 2000). Finally, I show that, following the FHAct, the homeownership rate and average net housing wealth increased at a faster pace among Black families than among White families.

Then, I simulate whether the Fair Housing Act was the driving force behind all of these changes. The end of lawful housing discrimination could have accelerated the outflow of Black families from Black neighborhoods, leading to decreased racial residential segregation. At the same time, better access to mortgages in more integrated neighborhoods (Aaronson et al., 2021) could have motivated more Black families to become homeowners and accumulate housing wealth. In this way, the Fair Housing Act could have led to decreasing racial segregation and declining racial differences in housing and total wealth in the long run. However, establishing such a causal relationship is challenging due to the ever-changing economic environment. Without further assumptions, it is impossible to distinguish the effect of the FHAct from other economic developments during the 1970s, like the rapidly growing real incomes of Black families.

To overcome this difficulty, I quantify the Fair Housing Act’s effect on residential segregation and racial wealth inequality within a quantitative model, which allows me to draw a causal relationship and rule out all economic changes of the 1970s other than the FHAct. I estimate a heterogeneous agent life-cycle overlapping generations model with spatial equilibrium and endogenous house prices. This quantitative model, among the first in the literature, allows me to jointly model the homeownership decision and location choices. In the model,

neighborhood-level house prices are set in spatial equilibrium, and households' relocation among neighborhoods changes house prices and, through them, housing wealth and wealth distribution. I estimate the model using the simulated method of moments to match the spatial and wealth distributions observed in the 1970 decennial census and the Survey of Consumer Finances Plus, assuming that some Black households were not allowed to move into White neighborhoods before the FHAct. As a validation exercise, I show that the estimated model generates realistic life-cycle homeownership and wealth-holding patterns for both Black and White families.

I simulate the effect of the Fair Housing Act by lifting relocation barriers in the model that prevented Black households from residing in White neighborhoods. As Black families relocate among neighborhoods in the model, house prices increase in neighborhoods that gain new residents and decline in those with declining populations, which could motivate White families to move as well. After the relocation of households, a new steady-state spatial equilibrium emerges with new house prices and new Black and White wealth distributions. I quantify the effect of the Fair Housing Act by comparing how the steady-state spatial distribution and wealth holding of households changed due to the relocation of Black families in the model.

According to my simulation results, the reduction of housing discrimination following the FHAct decreased the most common measure of segregation, the dissimilarity index between Blacks and non-Blacks, by 6.8 percentage points, which is slightly higher than the observed decline in segregation during the 1970s, which was 6.1 percentage points. I find that the FHAct reduced racial differences in average housing wealth and affected racial wealth inequality differently across the wealth distribution. In the model, the end of lawful housing discrimination moderately increased wealth inequality among Black families, as the wealth holding of the top 25% of the Black wealth distribution increased by 1.5-4%, while the wealth in the 6th Black wealth decile decreased by 1%. As the policy's effect on the White

wealth distribution was slightly positive between the 2nd and 8th deciles, racial differences in wealth holding decreased between the 8th and 10th deciles and increased between the 2nd and 7th deciles. My results suggest that reducing housing discrimination did not reduce racial wealth inequality across the entire wealth distribution; moreover, I find that racial wealth inequality increased in the majority of the wealth distribution as a result of the Fair Housing Act.

One of the main contributions of the paper is the joint modeling of households' homeownership and location choices within a dynamic model, yielding new insights into how homeownership decisions affect neighborhood choices over the life cycle. Aside from this paper, [Richard \(2024\)](#) studies how joint homeownership and location choice affect wealth inequality in a structural model. In this way, my paper contributes to the growing literature that studies how homeownership or spatial sorting affects wealth inequality.

My paper contributes to the literature on spatial sorting, in which both static ([Almagro and Domínguez-Iñó, 2024](#); [Weivu, 2024](#); [Davis et al., 2023](#)) and dynamic ([Fogli and Guerrieri, 2019](#); [Aliprantis and Carroll, 2018](#); [Chyn and Daruich, 2021](#)) models abstract from modeling homeownership. Because of this modeling simplification, these models cannot capture how families move between neighborhoods to accumulate housing wealth. I contribute to this literature by modeling homeownership decisions, race-dependent mortgage markets, and house market frictions directly, and by studying how ownership motivation affects Black and White families' location choices differently over their life-cycle. My paper also quantifies how the spatial relocation of households affects racial wealth inequality through housing wealth, a channel that has not been modeled directly in this literature.

My paper also contributes to the macroeconomics literature on the determinants of homeownership decisions over the life cycle ([Yang, 2009](#); [Brandsaas, 2021](#); [Khorunzhina and Miller, 2022](#)). By adding neighborhood choice to the usual dynamic framework in the literature, I quantify how the possibility of spatial relocation and changes in neighborhood-

level house prices affect the optimal timing of becoming a homeowner over the life cycle.

Finally, the paper contributes to the growing literature studying the determinants of Black–White wealth differences. Studies so far have investigated how racial differences in parental transfers ([Charles and Hurst, 2002](#); [Aliprantis et al., 2022](#)), housing and total wealth accumulation ([Boehm and Schlottmann, 2004](#); [Derenoncourt et al., 2024](#)), or family composition ([Ashman and Neumuller, 2020](#)) affect racial wealth inequality. However, the literature still lacks quantitative evidence on how the spatial sorting of families, and especially desegregation, affects wealth differences between Black and White families through the housing market. My paper contributes to this literature by showing that decreasing racial residential segregation has a sizable and heterogeneous effect on racial wealth inequality; my results suggest that desegregation could increase racial wealth inequality in the middle of the wealth distribution.

The paper proceeds as follows. In Section 2, I present three stylized facts about neighborhood sorting and changing trends in Black housing wealth accumulation during the 1970s. In Sections 3 and 4, I present and estimate a quantitative model that endogenously creates neighborhood sorting, changes in neighborhood-level house prices, and changes in the Black and White wealth distribution. In Section 5, I use this model to test whether the Fair Housing Act was the driving force behind the three presented stylized facts and to examine how the FHAct affected racial wealth inequality. Finally, I conclude my results in Section 6.

2 Stylized facts on segregation and racial wealth inequality during the 1970s

2.1 Data

The empirical analysis of this paper builds on two main datasets: (1) census tract-level aggregates from decennial censuses to track neighborhood-level changes over time, and (2) the Survey of Consumer Finances Plus (SCF+) to measure changes in household asset composition and wealth inequality. The presented spatial analysis is based on the Longitudinal Tract Data Base (LTDB) developed by [Logan et al. \(2014\)](#). The unit of observation in LTDB is census tracts, and the dataset contains census tract-level aggregates of population and various socioeconomic variables, starting with the 1970 decennial census. I used the 1970-2000 waves of the study and interpolated each census tract across different waves into 2010 census tract boundaries using LTDB's interpolation files. This interpolation allows me to track changes in time-invariant spatial units over time. I extended LTDB with aggregated census tract information from the 1960 census using the National Historical Geographic Information System (NHGIS) by interpolating the 1960 census tract boundaries into the 2010 tract definition.

Until 1990, census tract-level maps and tabulations were created only for core-based statistical areas (CBSA); thus, my sample only contains census tracts from metropolitan areas between 1960 and 2000. To focus on urban areas where Black/non-Black residential segregation was more prevalent, following [Reardon and Bischoff \(2011\)](#), I narrow the sample to census tracts in the 100 largest cities in 1970 and then keep only those with at least 10,000 Black individuals. The sample contains more than 36,000 census tracts from 87 metropolitan areas. Based on [Baum-Snow \(2007\)](#)'s neighborhood categorization, 23.4% of these tracts resided within the central city of the metropolitan area. Following [Ellen \(2000\)](#),

I categorized census tracts into three categories based on their share of the Black population at a given census year. I define a census tract as "White" if its Black share was less than 10%, as "Mixed" if its Black share was between 10-50%, and as "Black" if it was a majority Black neighborhood at a given census.¹

I use SCF+ to measure changes in homeownership, housing wealth, and racial wealth inequality. The SCF+ was developed first by [Kuhn et al. \(2020\)](#), and it is the most commonly used dataset to analyze racial wealth inequality in the second part of the 20th century ([Kuhn et al., 2020](#), [Derenoncourt et al., 2023](#); [Derenoncourt et al., 2024](#)). The SCF+ combines early waves of the Survey of Consumer Finances run by the Survey Research Center at the University of Michigan with the post-1983 waves of the study, which were run by the Federal Reserve System. The study was run each year between 1949-71, then every three years from 1989, while between 1971-89, it was run in 1977 and 1983. The study was redesigned in 1983, but [Kuhn et al. \(2020\)](#) harmonized variables and reweighed earlier waves to reflect the 1983 study design. SCF+ is the only dataset that contains both total family wealth and housing asset information starting with the early 1960s. However, as SCF+ lacks panel dimension and detailed location information, I cannot analyze how potential relocation affected Black families' wealth accumulation.

To control for the different survey sizes and irregularities in the frequency of SCF survey collection, for this analysis, I combine survey waves into 5-year bins. I follow [Derenoncourt et al. \(2024\)](#) by smoothing the time series of wealth components. While they take the unweighted moving average (MA3) of the time series, I smooth by bootstrapping the sample for a given bin from a pool that includes samples from neighboring bins as well; bootstrapping

¹Based on this categorization, majority Asian or Hispanic neighborhoods are also categorized as "White". The 1960 and 1970 censuses did not collect information on Hispanic origin, and the published white population of census tracts contains non-Hispanic and Hispanic whites as well. In these two censuses, the share of "White" neighborhoods with less than 50% of White population is just 0.1%. However, in line with the rising Hispanic population and the distinction between Hispanic and non-Hispanic whites in the 1980-2000 censuses, the share of "White" neighborhoods with less than 50% of the non-Hispanic white population in the 1980-2000 censuses was 5.2%, 9.4%, and 16.2%, respectively.

also allows me to calculate confidence intervals for wealth measures. Every wealth component is measured in 1970 prices, which I deflated using the Census Bureau’s C-CPI-U index.

2.2 Stylized facts

Stylized fact I: The 1970s was a turning point in the history of segregation in the US

The Fair Housing Act in 1968 and the Equal Credit Opportunity Act in 1974 eased the legal barriers that Black families faced when moving into majority-White neighborhoods, which dramatically changed the landscape of segregation in the United States (Boustan, 2011). Table 1 shows that the national weighted average of the most often used measure of segregation, the dissimilarity index between Black and non-Black individuals, peaked at the 1970 decennial census and then declined rapidly in the following decades.² In just three decades, the average of the CBSA-level dissimilarity indexes dropped by 18.6 percentage points. Based on Cutler et al. (1999)’s calculation, the magnitude of this decline is larger than the dissimilarity index’s increase during the Great Migration (17.7 percentage points increase between 1910-70) when nearly six million Black people moved from Southern rural areas to Northern urban centers. As a result, by 2000, the national dissimilarity index dropped just under the threshold of 0.6, above which cities are considered very segregated (Boustan, 2011).

The extensive literature on racial segregation suggests that the decline of racial segregation was the result of the widespread relocation of Black families from majority Black neighborhoods into more integrated ones (Ellen, 2000; Fischer et al., 2004). As the spatial distribution of Black agents in Table 1 illustrates, the fraction of Black individuals living in

²The dissimilarity index measures how segregated a minority group’s spatial distribution is from a majority population within a larger spatial unit (like a metropolitan area). Its value can range from 0 to 1, with larger values indicating a higher level of segregation. Intuitively, the dissimilarity index indicates the fraction of the minority group that should be relocated to other neighborhoods to achieve a random distribution of the minority group. See Massey and Denton (1988) or Reardon and Bischoff (2011) for more details.

Table 1: Segregation measures and distribution of families by Black share of neighborhoods

	1960	1970	1980	1990	2000
Residential segregation (Dissimilarity index)					
Black vs. Non-Black	0.780%	0.785%	0.724%	0.657%	0.599%
Share of Black population in White neighborhoods					
Blacks / All population	1.07%	1.16%	1.89%	2.46%	3.03%
Spatial distribution of Black population					
White neigh.	7.5%	7.2%	10.2%	12.6%	13.4%
Mixed neigh.	25.7%	22.1%	23.3%	28.1%	33.3%
Black neigh.	66.8%	70.7%	66.6%	59.3%	53.3%
Spatial distribution of Non-Black population					
White neigh.	87.1%	87.7%	85.3%	81.6%	76.9%
Mixed neigh.	10.7%	10.1%	12.3%	16.0%	20.6%
Black neigh.	2.2%	2.2%	2.4%	2.4%	2.5%
Income segregation (Rank order H), from Reardon et al. (2011)					
Among Black families		0.099	0.133	0.173	0.170
Among White families		0.110	0.117	0.132	0.139
All families		0.124	0.134	0.152	0.157

Sources and notes: 1. Dissimilarity index: own calculation based on LTDB and NHGIS data. The overall dissimilarity index is calculated as the population-weighted mean of the dissimilarity index of the 87 CBSAs in the sample. 2. Spatial distribution of Black population: own calculation based on LTDB and NHGIS data. Averages are weighted by tract population. Neighborhoods are defined based on the Black share at each census year. 3. Income segregation information is taken from [Reardon and Bischoff \(2011\)](#).

Black neighborhoods increased between 1960-70, but then it started to decline continuously. Within 30 years, the fraction of Black individuals living in White neighborhoods almost doubled, as it increased from 7.2% to 13.4%. Meanwhile, another measure of Black presence in White neighborhoods, the average share of Black individuals in White neighborhoods nearly tripled, from 1.16% to 3.03%. By 2000, 30 years after the Fair Housing Act, almost as many Blacks lived in White and Mixed neighborhoods as in the majority Black ones. Meanwhile, the fraction of non-Black individuals living in White neighborhoods started to decline during the 1970s, primarily because many White tracts became Mixed due to the inflow of Black families.

In line with the decreasing racial residential segregation, the income residential segregation started to increase during the 1970s. As [Table 1](#) shows, income segregation increased within both Black and White families, but the rise of income segregation was much faster

Table 2: Change of average Black and non-Black population in Black neighborhoods

	1960	1970	1980	1990	2000
Panel A: Based on 1970 neighborhood categorization					
# of Black	2,870	3,774	3,180	2,731	2,429
# of non-Black	1,992	795	550	658	808
Total	4,862	4,568	3,730	3,389	3,237
Change of Black population		31.5%	-15.7%	-14.1%	-11.1%
Panel B: Based on 1960 neighborhood categorization					
# of Black	4,039	3,868	3,013	2,553	2,274
# of non-Black	1,038	579	547	696	870
Total	5,077	4,446	3,560	3,248	3,145
Change of Black population		-4.2%	-22.1%	-15.3%	-10.9%
Panel C: Based on changing neighborhood categorization					
# of Black	4,039	3,774	3,185	2,859	2,749
# of non-Black	1,038	795	680	674	706
Total	5,077	4,568	3,865	3,533	3,455
Change of Black population		-6.6%	-15.6%	-10.2%	-3.8%

Sources and notes: 1. Spatial distribution of Black population: own calculation based on LTDB and NHGIS data. The sample contains census tracts in the 100 largest CBSAs, which had at least 10,000 Black residents in 1970.

among Blacks ([Reardon and Bischoff, 2011](#)). The rapidly increasing income segregation among Black families suggests that most of the relocating families were from the middle and top of the Black income distribution, many of whom were potentially not allowed to move into majority White neighborhoods before the Fair Housing Act ([Hirsch, 1983](#)).

Stylized fact II: Black families started to leave Black neighborhoods during the 1970s

The relocation of Black families from majority Black neighborhoods had a sizable effect on these neighborhoods' population, and it likely sped up the population decline of Black neighborhoods. [Table 2](#) summarizes how the average number of Black and non-Black residents changed over time in Black neighborhoods.³ In [Table 2](#), I use three different neighborhood categorizations: I categorized neighborhoods by their Black share in 1970 in Panel A, based on their Black share in 1960 in Panel B, and I defined neighborhoods based on

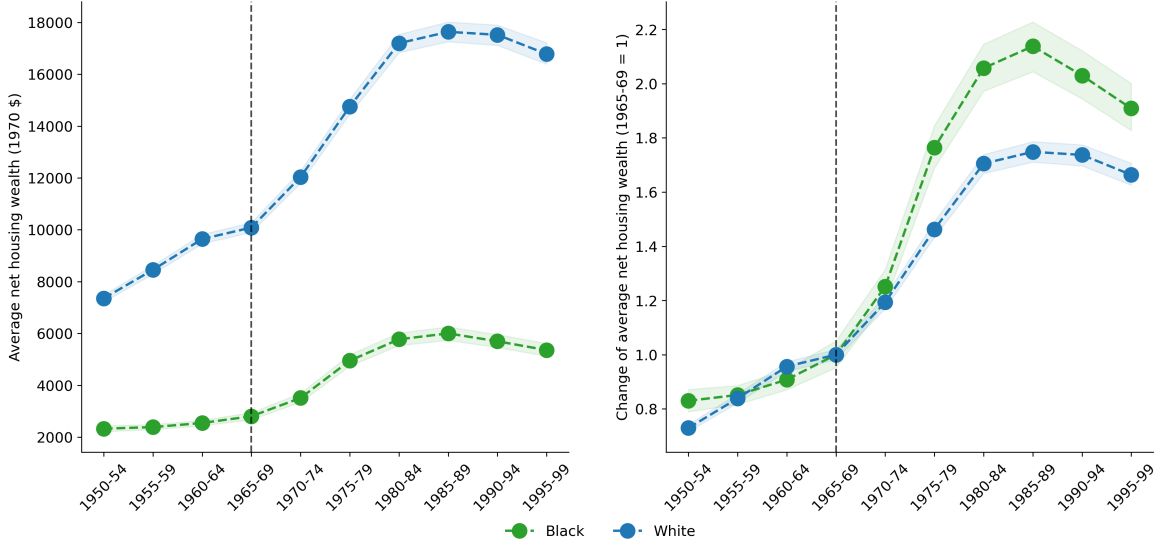
³In [Table 2](#), I present the number of non-Black rather than non-Hispanic White individuals, as computing the latter is not possible for the 1960 and 1970 censuses (due to the lack of information on Hispanic origin).

their Black share of each census year in Panel C. Compared to Panel A and B, Panel C contains neighborhoods which became majority Black after 1960 or 1970, and thus the set of neighborhoods in Panel C changes between census years. Panel A shows how neighborhood composition changed in tracts that were majority Black right after the Fair Housing Act, while Panels B and C provide a robustness check by showing that the population change in Panel A is not the consequence of the choice of neighborhood definition.

Panel A of [Table 2](#) shows that, on average, neighborhoods that had more than 50% Black share in 1970 experienced a large inflow of Black individuals between 1960-70. This trend reversed during the 1970s when the average number of Blacks in these tracts declined by 15.7%, and outmigration of Black families continued from Black neighborhoods in the following decades as well. By lowering the legal obstacles Black families faced during relocation, the Fair Housing Act could have supported this sudden change in Black neighborhoods during the 1970s.

The large number of non-Black individuals who left Black neighborhoods during the 1960s suggests that many of these neighborhoods experienced a quick change in their racial composition, which, in theory, could motivate Black families to leave these tracts. In general, it could be possible that Black neighborhoods experienced a large outflow of the Black population right after they became majority Blacks, which could also explain population changes in Panel A. However, as Panel B of [Table 2](#) shows, counter to this explanation, neighborhoods that were already majority Black in 1960 experienced only a modest decline in Black population during the 1960s, but they went through similar depopulation during the 1970s as neighborhoods in Panel A. Panel C of [Table 2](#), in which the sample of tracts contains all neighborhoods which were majority Black at the given census year, draws a similar picture by showing that the average number of Black population in Black neighborhoods declined the most in the 1970s. Panel B and C suggest that the large decrease in the number of Black population in Black neighborhoods between 1970-80 was a result of a unique change

Figure 1: Evolution of average net housing wealth



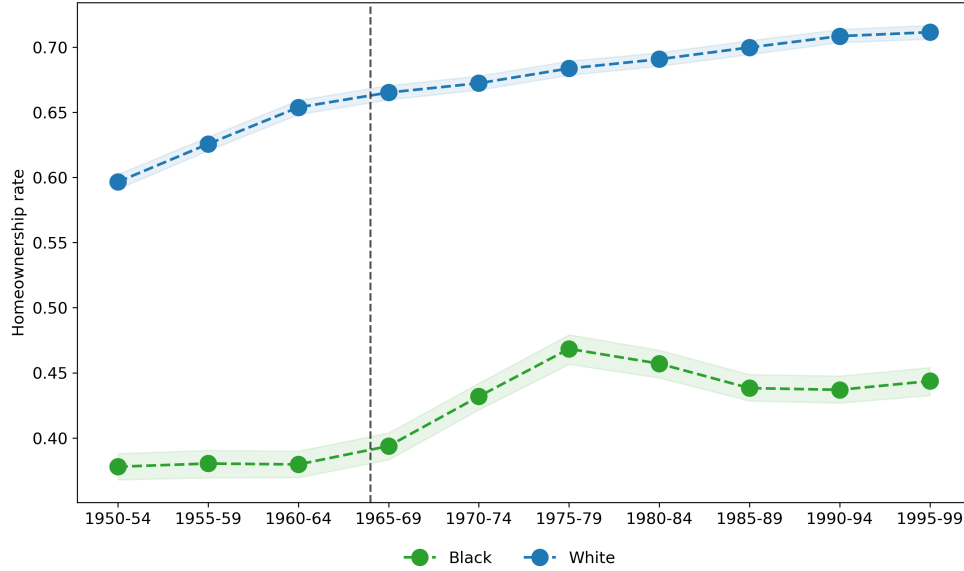
Source: own calculation, based on SCF+ data. Housing wealth is defined as asset value minus the remaining mortgage of the first residence. Transparent areas represent bootstrapped 95% confidence intervals.

in relocation behavior during the 1970s.

Overall, all three neighborhood definitions show that Black neighborhoods experienced a double-digit decline in the Black population during the 1970s, and that the decline continued but slowed in subsequent decades. During the same time span, the Black population of the United States increased by 15-20% in each decade between 1960 and 2000, which makes the depopulation of Black neighborhoods even more dramatic. Interestingly, despite the declining Black population in these tracts, the decline of the average non-Black population reversed between 1980 and 1990, and Black neighborhoods experienced increasing non-Black presence during the 1990s.

Stylized fact III: Black families' housing wealth increased at a faster pace than Whites'

Figure 2: Black and White homeownership rate

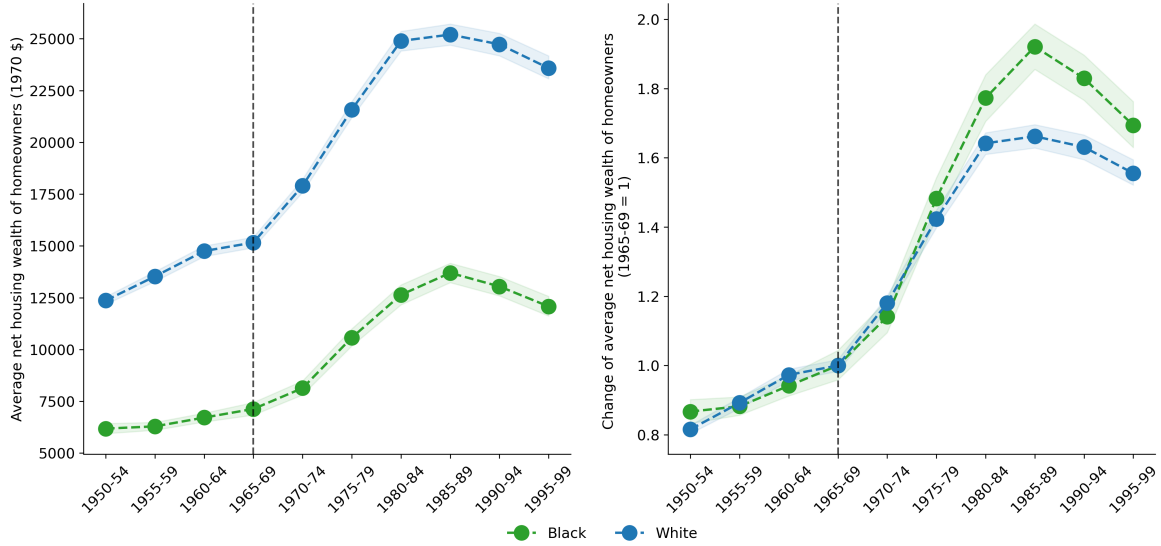


Source: own calculation, based on SCF+ data. Transparent areas represent bootstrapped 95% confidence intervals.

Differences in Black and White housing wealth have always been at the center of discussions about racial wealth inequality (eg. [Shapiro, 2017](#); [Rothstein, 2018](#)). As housing assets dominate the asset portfolios of families in the middle of the wealth distribution ([Aladangady and Forde, 2021](#)), racial differences in housing wealth accumulation could create sizable differences in racial wealth inequality over time ([Charles and Hurst, 2002](#)). Meanwhile, net housing wealth is the wealth component most likely affected directly by the Fair Housing Act. As the right panel of [Figure A1](#) in the Appendix shows, the total net wealth of Black and White families increased at the same pace during the second part of the 20th century, while total non-house-related wealth increased faster among Whites than Blacks ([Figure A2](#)). However, as the right panel of [Figure 1](#) shows, average net housing wealth increased much faster among Black families than among Whites after the Fair Housing Act.

The left panel of [Figure 1](#) illustrates that average White housing wealth rose rapidly between 1950 and 1980, when the nominal difference in average housing wealth between Black and White families increased. Compared to Whites, Black families' housing wealth

Figure 3: Evolution of average net housing wealth of homeowners



Source: own calculation, based on SCF+ data. Housing wealth is defined as asset value minus the remaining mortgage of the first residence. Prices measured in 1970 dollars. Transparent areas represent bootstrapped 95% confidence intervals.

accumulation was much slower until the 1970s, when it started to accelerate. The right panel of Figure 1, which shows how the average housing wealth changed compared to its 1965-69 value, indicates that during the 1970s, the average housing wealth of Black families increased at a faster pace than Whites families', which was not the case in previous decades. As the narrowing of nominal differences between Black and White average housing wealth requires faster housing wealth accumulation by Black families, Black families' larger housing wealth growth than Whites' in the 1970s could have been a necessary first step toward the long-run decline in racial differences in housing wealth. Thus, despite the steadily increasing nominal differences in average housing wealth between Black and White households, the 1970s seemed to be a turning point in Black families' housing wealth accumulation.

As average net housing wealth is the product of the homeownership rate and the average net housing wealth of homeowners, we can decompose whether the rapid rise in average Black net housing wealth was driven by increases in homeownership, increases in homeowners' net housing wealth, or both. Figure 2 shows that Black homeownership rose rapidly during the

1970s but then declined somewhat, which could partly explain the slowdown in the average increase in Black housing wealth in the 1980s. Nevertheless, the expansion of the Black homeownership rate during the 1970s was the largest of any decade in the second half of the 20th century. White families' homeownership rate increased steadily at a similar pace between 1960 and 2000, with no noticeable change during the 1970s.

Meanwhile, as [Figure 3](#) depicts, during the 1970s, Black homeowners accumulated housing wealth at a faster pace than in earlier decades. During the 1970s, the growth rate of average housing wealth of Black homeowners was nearly the same as that of Whites (right panel of [Figure 3](#)), and the nominal difference between the average housing wealth of Black and White homeowners continuously increased (left panel of [Figure 3](#)).

As a result, the increase in nominal Black housing wealth in the 1970s (left panel of [Figure 1](#)) was a byproduct of the rapidly growing Black homeownership rate and the acceleration in the growth of Black homeowners' housing wealth. However, as Black homeowners acquired housing wealth at the same pace as White homeowners, the faster accumulation of housing wealth by Black families during the 1970s (right panel of [Figure 1](#)) was driven by declining racial differences in homeownership rates.

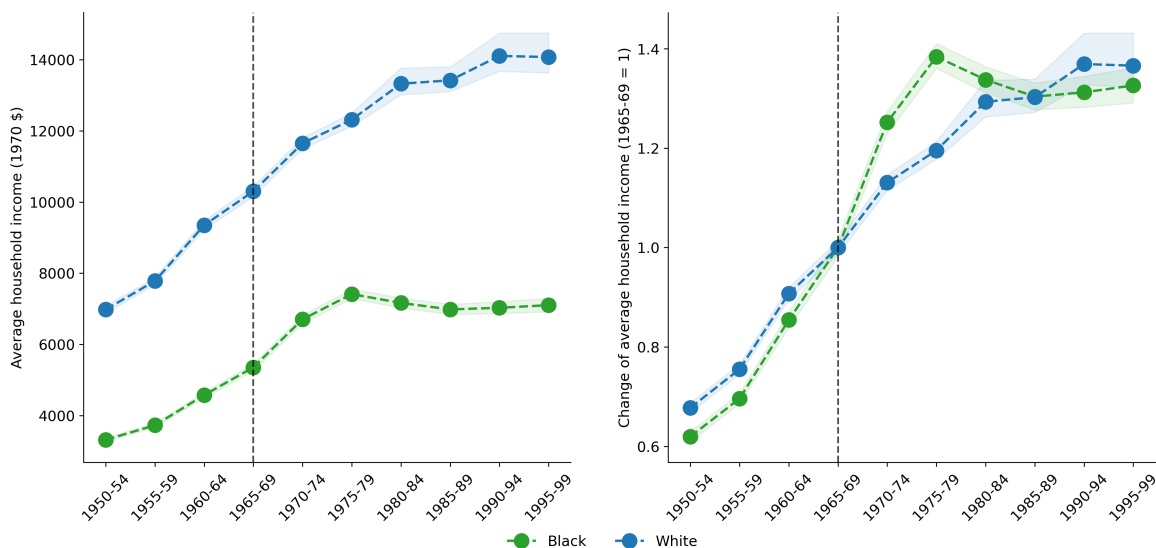
In summary, the 1970s were a favorable environment for the accumulation of housing wealth among Black families. Because of existing racial differences, the nominal gap between average Black and White housing wealth widened during this time. However, Black families accumulated housing wealth at a faster pace than Whites during the 1970s, which could have been the necessary first step toward closing existing racial differences in housing wealth.

Can we draw a causal line between the FHAct and these stylized facts?

The 1970s were a turning point in both the history of racial segregation and racial wealth inequality in the United States. The decades-long increase in racial segregation reversed, and since 1970, the Black/non-Black racial segregation has been in constant decline. During the

same time, Black families started to move out at an accelerated pace from majority Black neighborhoods, and Black families experienced a quickly increasing homeownership rate and faster housing wealth accumulation than Whites.

Figure 4: Evolution of average household income



Source: own calculation, based on SCF+ data. Prices measured in 1970 dollars. Transparent areas represent bootstrapped 95% confidence intervals.

One of the key questions in this paper is whether the Fair Housing Act could explain all of these three developments within a single causal framework. By easing housing discrimination against Black families, the Fair Housing Act could have caused the sizable relocation of these families to more integrated neighborhoods (Stylized Fact II), which could explain the declining residential segregation (Stylized Fact I). Meanwhile, easier access to mortgages in more White neighborhoods could have motivated Black families to move there, become homeowners, and begin accumulating housing wealth (Stylized Fact III).

However, drawing a causal relationship among the FHAct, these three stylized facts, and the evolution of racial wealth inequality would be questionable given the changing economic environment over these decades. As Figure 4 shows, the total income of Black families grew faster than that of White families during the 1970s, which likely contributed to the

quickly increasing homeownership and housing wealth of Black families. Without further assumptions, we cannot distinguish the positive effect of increasing average Black household income on racial wealth inequality from the potential effect of declining housing discrimination following the Fair Housing Act. Because of these difficulties, I investigate the potential effects of the Fair Housing Act on these three stylized facts and on racial wealth inequality using a structural model, which enables me to quantify the effect of relocating Black families on racial wealth inequality while holding all other relevant economic conditions (like income) constant.

3 Model

In this section, I introduce a quantitative structural model to quantify how the relocation of Black families affected the spatial sorting of Black and White households and how this relocation altered Black-White wealth differences at different parts of the wealth distribution.

The model builds on a life-cycle heterogeneous agent OLG framework (similarly to [De Nardi, 2004](#); [Ashman and Neumuller, 2020](#); [Brandsaas, 2021](#); or [Chyn and Daruich, 2021](#)), and generates wealth inequality through differences in lifetime income and hand-to-mouth behavior. Agents in the model can endogenously relocate across three neighborhoods, which I estimate to represent White, Mixed, and Black neighborhoods. Unlike most models in the literature with spatial equilibrium (like [Aliprantis and Carroll, 2018](#); [Fogli and Guerrieri, 2019](#); or [Chyn and Daruich, 2021](#)), I model homeownership decisions directly, which allows me to model homeownership and location choice jointly and investigate whether the Fair Housing Act affected renters and homeowners differently. House prices and rents in neighborhoods are endogenous and set in equilibrium, which generates neighborhood-level house price changes in response to population change. As in [Aliprantis and Carroll \(2018\)](#), I assume that some Black families are initially not allowed to move into the model’s White neighborhood.

3.1 Environment

Demographics The model is populated with overlapping generations of agents, who can be either Black or White ($R = \{B, W\}$, where R denotes race). Each period in the model can be interpreted as one year of life. Agents enter the model at age 24, in the neighborhood where their grandparents resided. New agents inherit their parents' race, their time-invariant and race-dependent income fixed effect ρ_R , and a bequest from their grandparents. Agents have one child, who enters the model when agents are 48 years old. Agents retire at age 60, after which they receive a time-invariant pension income; otherwise, their decision problem does not change. Agents leave the model after age 72, when they can leave a bequest to their grandchildren, who inherit it in the next period when they enter the model.

Preferences In each period, agents gain utility from consuming consumption goods (c) and housing services ($v(h)$) and from neighborhood amenities by living in the neighborhood n :

$$u(c, h, n)_R = \frac{(c^{1-\alpha}v(h_n)^\alpha)^{1-\sigma}}{1-\sigma} + \kappa_{R,n}A_n,$$

where α is the expenditure shares of housing consumption, σ measures risk aversion, and A_n is a race-independent amenity level that represents the utility derived from living in neighborhood n . Parameter $\kappa_{R,n}$ is a race-dependent taste factor for living in neighborhood n , which allows Black and White agents to gain different utility by living in the same neighborhood. As a normalization, I set all three of White agents' $\kappa_{W,n}$ to 1. Just like in [Brandsaas \(2021\)](#), function $v(\cdot)$ maps consumption of housing good in the neighborhood n to utility

$$v(h_n) = \begin{cases} \eta_{n,rent} & \text{if renting in neighborhood } n \\ \chi\eta_{n,own} & \text{if owning in neighborhood } n \quad (\chi > 1) \end{cases}$$

where χ represents the additional utility derived from owning a house (like potential higher-quality or other benefits that are not explicitly modeled, such as tax benefits). In the model homeownership h is binary ($h = 1$ if the agent is a homeowner, and $h = 0$ if the agent is a renter), but the consumption value of housing good $v(h_n)$ is non-binary, and it differs among neighborhoods. At age 72, agents also derive utility from leaving non-negative bequests to their grandchildren, which, following [De Nardi \(2004\)](#), takes the form of

$$beq(b', h', n') = \theta_{1,R} \left(1 + \frac{(1+i)b' + p'_n h' (1-\delta)}{\theta_2} \right)^{1-\sigma} \quad \text{where } b' \geq 0$$

where the race-dependent $\theta_{1,R}$ captures the parents' concern about leaving the bequest to their children, and the size of θ_2 controls the extent to which the bequest is a luxury good. The nominator represents the wealth inherited by offspring: b is bond holding, i is the risk-free interest rate of bonds, p_n is the neighborhood-specific house price, h is an indicator for owning a house, and δ is the per-period depreciation of housing goods.

Bequest connects generations in the model and allows agents to hold more wealth than would be possible just by their own income. In the model $\theta_{1,R}$, which controls how much utility agents gain by leaving bequests, is race-dependent for two reasons. First, the SCF+ shows significant racial differences in wealth at age 24 when agents enter the model: the median wealth holding of Black and White households at that age is \$0 and \$1,224, respectively. Second, previous research ([Charles and Hurst, 2002](#); [Ashman and Neumuller, 2020](#)) highlighted that racial differences in parental involvement in down payments are an important source of the racial differences in the homeownership rate. As there are no in-vivo transfers in the model, racially differing bequests capture the existing racial differences in parental help in becoming a homeowner.

Neighborhoods There are three neighborhoods n in the model: Black, Mixed, and White.

Neighborhoods differ in endogenous house prices p_n and rents q_n , in exogenous amenity levels A_n , and in consumption utility derived from owning $\eta_{n,own}$ or renting $\eta_{n,rent}$ in the neighborhood. Agents moving from neighborhood n to n' need to pay a relocation cost:

$$RC(n, n') = \begin{cases} k & \text{if } n \neq n' \\ 0 & \text{if } n = n' \end{cases}$$

I model the effect of the Fair Housing Act by imposing an exogenous relocation constraint F that affects only Black agents. Agents with this constraint ($F = 1$) cannot move into the White neighborhood, while Black agents without this constraint ($F = 0$) – just as White agents – can choose from any neighborhoods to live in. I assume that π_F fraction of Black households living in the Black or Mixed neighborhood have this relocation constraint.

Housing market Agents can either rent or own a house. Neighborhood-level house prices are set in equilibrium by a supply function $S_n = \bar{S}_n p_n^\Delta$, where p_n is the equilibrium house price in neighborhood n , Δ is the price elasticity of housing supply, and $\bar{S}_n$ is a supply shifter. Housing assets depreciate between periods with a δ rate, and homeowners need to pay selling and buying costs $SBC(n, n')$ if their housing stock changes between periods:

$$SBC(h, h') = \begin{cases} e_s p_n & \text{if } h' < h \\ e_b p_n & \text{if } h' > h \\ 0 & \text{if } h' = h \text{ \& } n' = n \end{cases}$$

where e_s and e_b are selling and buying costs, modeled as a fraction of the price. Agents with housing wealth do not need to pay selling or buying costs if they keep their house, but they need to pay both selling and buying costs in case they move into a different neighborhood

and buy a house again. Households without housing assets need to pay rent q_n , which is a function of house price p_n .

Income process Active agents between the ages of 24-60 provide one unit of labor supply inelastically in each period. Their income endowment depends on their age, their race-dependent permanent income fixed effect ρ_R , and their race-dependent stochastic income shock ϵ_{ρ_R} . Retired agents between the ages of 60-72 receive a pension, which is a constant fraction of their pre-retirement wage. Agents need to pay progressive income tax $tax(y)$.

Financial market Each period agents can save in bonds b , which pays $(1 + i)b$ at the beginning of the next period. Agents can have negative bond holding to buy a house if they receive a mortgage, in which case agents owe $(1 + i^m)b$ at the beginning of the next period. Agents cannot hold mortgages and savings in bonds at the same time, and the rate of return on risk-free bonds is lower than the mortgage interest rate ($i < i^m$). The probability of receiving a mortgage in a given period is stochastic, and it depends on the income and race of the agent. At the beginning of each period, agents learn whether they can receive a mortgage ($\mu = 1$) or not ($\mu = 0$). To receive a mortgage, agents also need to satisfy an age-dependent debt-to-income (DTI) and loan-to-value (LTV) ratio, which means agents need to have enough savings first to be able to buy a house. Agents holding a mortgage are guaranteed to have access to a mortgage at the next period, but the actual amount of the next period's mortgage depends on their stochastic income at the next period. Agents need to pay back all of their mortgage loans at age 72, at the latest.

Hand-to-mouth (HtM) households To match the reality that a high fraction of Black households had wealth holding less than 2 weeks of their monthly income in 1970, the model contains hand-to-mouth agents, which is a time-invariant and inheritable trait. The only

difference between hand-to-mouth ($HtM = 1$) and non-hand-to-mouth ($HtM = 0$) agents is their discount factor, which is lower for hand-to-mouth households; otherwise, hand-to-mouth agents face exactly the same decision problem as non-hand-to-mouth households.

3.2 Homeowners' recursive problem

To make households' decision problem more tractable, each variable without a comma (eg. n or age) represents a state variable, while variables with commas can be either a choice variable in the given period (c' is the optimal consumption in the given state) or next period state variables (eg. age'). I define homeowners as agents who decide to own a house in a given period ($h' = 1$), independently of their initial holding of housing units (h). Depending on agents' initial bond holding (b) and homeownership (h), their residence (n), their stochastic income shock (ϵ), and the possibility of receiving mortgage (μ), in each period agents decide how much to consume (c'), save in bonds (b'), whether to own a house (h') and where to live (n'). I model agents' decision problem in two steps: first, for each neighborhood, agents make their expected lifetime-utility maximizing consumption, saving, and house purchasing decision, and then they choose the utility-maximizing neighborhoods to live in. We can represent the recursive problem of an agent with race R , permanent income fixed effect ρ , hand-to-mouth condition HtM , and relocation barrier F who decides to own a house in a given period with the following equations:

$$V^{R,\rho,F,HtM}(age, b, h, n, \epsilon, \mu) = \max_{c', b', h', n'} \left\{ u(c', h', n') + \beta_{HtM} \mathbb{E} [V^{R,\rho,F,HtM}(age', b', h' = 1, n', \epsilon', \mu') | \epsilon, \mu] \right\}$$

st.

$$y - tax(y) + b(1 + i^*) + p_n h(1 - \delta) = c' + p_{n'} h' + b' + RC(n, n') + SBC(h, h' = 1)$$

$$b' \geq \begin{cases} 0 & \text{if } \mu = 0 \text{ or } age' > 72 \\ \max\{-LTV \times p_n h' ; -DTI \times y \times (1 - (1 + i^m)^{-24})/i^m\} & \text{if } \mu = 1 \text{ \& } age' \leq 72 \end{cases}$$

$$i^* = \begin{cases} i & \text{if } b \geq 0 \\ i^m & \text{if } b < 0 \end{cases}$$

$$V^{R,\rho,F}(age' > 72, b', h' = 1, n', \epsilon, \mu) = \theta_{1,R} \left(1 + \frac{(1+r)b' + p_n h' (1-\delta)}{\theta_2} \right)^{1-\sigma}$$

3.3 Renters' recursive problem

Renters are defined as agents who decide to not hold housing wealth in a given period ($h' = 0$). Renters differ from homeowners in two important dimensions. First, these households need to pay rent q_n instead of living in their own house, which also means they do not transfer wealth across periods in housing wealth ($h' = 0$). Second, as borrowing is allowed in the economy only to finance a house purchase, they cannot choose a negative bond amount ($b' \geq 0$). Then, we can represent the recursive problem of an agent with race R , permanent income fixed effect ρ , hand-to-mouth condition HtM , and relocation barrier F who decides to rent in a given period with the following equations:

$$V^{R,\rho,F,HtM}(age, b, h, n, \epsilon, \mu) = \max_{c', b', h', n'} \left\{ u(c', h', n') + \beta_{HtM} \mathbb{E} [V^{R,\rho,F,HtM}(age', b', h' = 0, n', \epsilon', \mu') | \epsilon, \mu] \right\}$$

st.

$$y - tax(y) + b(1 + i^*) + p_n h(1 - \delta) = c' + q_{n'} + b' + RC(n, n') + SBC(h, h' = 0)$$

$$b' \geq 0$$

$$i^* = \begin{cases} i & \text{if } b \geq 0 \\ i^m & \text{if } b < 0 \end{cases}$$

$$V^{R,\rho,F}(age' > 72, b', h' = 0, n', \epsilon, \mu) = \theta_{1,R} \left(1 + \frac{(1+r)b'}{\theta_2} \right)^{1-\sigma}$$

3.4 Stationary equilibrium

For a given house price vector P , a stationary equilibrium can be characterized with a collection of value functions $V^{R,\rho,F,HtM}(age, b, h, n, \epsilon, \mu)$, a set of policy functions $c'^{R,\rho,F,HtM}(age, b, h, n, \epsilon, \mu)$, $b'^{R,\rho,F,HtM}(age, b, h, n, \epsilon, \mu)$, $h'^{R,\rho,F,HtM}(age, b, h, n, \epsilon, \mu)$, and $n'^{R,\rho,F,HtM}(age, b, h, n, \epsilon, \mu)$, and distribution of households $\Omega_{age}(b, h, n, \epsilon, \mu, R, \rho, F, HtM)$, which satisfies the following conditions:

1. For house price vector P , the policy functions maximize all agents' expected lifetime utility.
2. Policy functions ensure spatial equilibrium, at which the housing market clears in each neighborhood at price vector P .
3. The state distribution of each cohort at any age is invariant over time.

4 Estimation

I estimate the model to match the spatial and wealth distribution of Black and White households observed in 1970. I interpret this steady state of the model as the “pre-segregation” era, assuming that most of the relocation of Black families, and thus the decline in residential segregation, took place after 1970, and not between 1968 (when the Fair Housing Act was

enacted) and 1970. I make this assumption to be able to compute the spatial distribution of families using the 1970 decennial census. This assumption seems reasonable based on [Table 2](#), which shows that large-scale relocation of Black families from Black neighborhoods began in the 1970s rather than the 1960s.

During estimation, I used the census-tract-level aggregated 1970 Census data (LTDB and NHGIS), the 1969-71 waves of SCF+, and the 1968-74 waves of the Panel Study of Income Dynamics (PSID). As my census data contains only urban census tracts, I narrowed the SCF+ sample to families residing in metropolitan areas whose heads were between the ages of 24 and 72. All calculations are weighted using sample weights (SCF+ and PSID) or the total population of tracts (1970 Census).

4.1 Parameters chosen outside of the model

[Table 3](#) summarizes all relevant parameters estimated outside of the model.

Preferences Following the literature, I set the risk aversion parameter γ to 2, the discount factors $\beta_{HtM=0}$ to 0.96, and $\beta_{HtM=1}$ to 0.72 ([Aguiar et al., 2020](#)), and the expenditure share of consumption good α to 0.875 ([Brandsaas, 2021](#)). To simplify the estimation process, I set θ_2 parameter to 100. Similarly, I set the ratio of White hand-to-mouth agents to 12.14%, which is the fraction of White households with net wealth holding less than 2 weeks of income in the SCF+.

Income process I model the income process with three components. First, a deterministic age-dependent part ensures that household income follows a life-cycle trajectory observed in the data. Second, a time-invariant income fixed effect generates lifetime income inequality among households, which allows higher-income households to accumulate more wealth over time. Finally, a stochastic income shock term captures individual-level income shocks, which

generate income uncertainty during working age. Estimating the latter two components requires panel dimension, which, unfortunately, is lacking from the SCF+ dataset, which I use to target moments during estimation. To overcome this barrier, I estimate the income process in two steps.

First, using PSID, I estimate the following equation separately for Black and White households

$$\log(y_{i,t}) = \beta_0 + \beta_1 age_{i,t} + \beta_{2,R} age_{i,t}^2 + \beta_{3,R} age_{i,t}^3 + FE_i + e_{i,t}$$

where $y_{i,t}$ is total household income inflated into 1970 prices, the cubic polynomial of age is the life-cycle trajectory of the income endowment, FE_i is a fixed effect capturing the permanent income fixed effect, and $e_{i,t}$ is an error term. To calibrate the model's income fixed effect ρ_R , first I divide the sample into three terciles based on the predicted $\widehat{FE_i}$ parameters, and then I set the vector of ρ_R as the weighted mean of the predicted $\widehat{FE_i}$ parameters in these terciles. Then, I calibrate the temporary income shock parameter ϵ_{ρ_R} conditionally on ρ_R : within each $\widehat{FE_i}$ terciles I create three new tertiles based on $\widehat{e_{i,t}}$, and define the vector of ϵ_{ρ_R} as the weighted mean of $\widehat{e_{i,t}}$ in these new terciles. I assume that ϵ_{ρ_R} follows a Markov process with identical transition probabilities (1/3). Calibrating ϵ_{ρ_R} conditionally on ρ_R allows me to set heterogeneous income shocks by income fixed effect and race, which creates more realistic temporary shocks and income endowments. In the end, this calibration exercise yields three ρ_R and nine ϵ_{ρ_R} values for each race.

In the second step, using the SCF+ data, I regress household income on a cubic polynomial of age. I use the predicted parameters from this regression in the model instead of those estimated using PSID because they better fit the Black and White income distribution observed in SCF+. After retirement at age 61, agents receive a constant fraction of their age 60 income as a pension, with fractions of 60.7% for Black and 70.6% for White households, estimated using the PSID.

I calculated income taxes using the 1970 income tax brackets of married couples filing jointly, which contained 25 marginal tax rates; I set the standard deduction to \$625. To ensure that all households can afford to pay rent, I set the after-tax minimum income in the model to the rent in the White neighborhood. Appendix B contains the estimated income parameters, the overall fit of the income processes, and the income tax brackets.

Mortgage market As [Ladd \(1998\)](#) shows, racial mortgage market discrimination was widespread during the 1960s and 1970s, but based on the 1970 Census of Housing ([U.S. Bureau of the Census, 1973](#)), the discrimination seemed to be more prevalent in the extensive margin (whether to get a mortgage) than in the intensive margin (mortgage conditions). The 1970 Census of Housing shows that mortgage conditions of Black applicants were quite similar to population averages in interest rate (6.1% vs. 6.0%), in duration (23.9 vs. 24.5 years), in debt-to-income ratio (11% to 9%), and in loan-to-value ratio at buying (86% vs. 78%, for conventional mortgages). Because of these similarities, I assume that Black and White households face the same mortgage conditions in the model, but the probability of receiving a mortgage differs by race.

I calculated the debt-to-income (DTI) ratio using the 1968-74 waves of the PSID, as SCF+ doesn't include information on annual mortgage payments. The weighted average of DTI for new purchases was 14.46%. I set the mortgage length as 24 years based on the 1970 Census of Housing. These two conditions, together with the internally estimated mortgage rate, define the maximum per-period annuity mortgage payment, from which I can calculate the maximum mortgage amount in the model. The LTV ratio also imposes an additional restriction on the maximum mortgage amount, which I calculated using the SCF+. In the model, I assumed that LTV is age-dependent and calculated the LTV ratios as the predicted cubic polynomial in age (as illustrated in [Figure A4](#)). Because of this assumption, agents cannot hold a high LTV ratio over their lifetime, which makes their life-cycle saving decisions

more realistic. I assume that the probability of receiving a mortgage is in the form of

$$P_{i,R}(\mu' = 1|\mu = 0) = \min\left(\frac{y_i}{y_{75}}\psi_R; 1\right); \quad P_{i,R}(\mu' = 1|\mu = 1) = 1$$

where y_i is an agent's income, y_{75} is the 75th percentile of household income from SCF+, and ψ_R is a race-dependent discount factor. This functional form guarantees that agents with mortgages will have access to mortgages in the future, that the probability of getting a mortgage increases with income, and that for most income levels, Black agents are less likely to receive mortgages. Using mortgage application data from the late 1970s, [Schafer and Ladd \(1981\)](#) found that Black applicants were 1.58–7.82 times as likely to be rejected as Whites with similar observable characteristics. Using this study I set $\psi_B = 1/1.58 = 63.29\%$ for Black agents, while as a normalization I define $\psi_W = 1$ for Whites.

Neighborhoods I use the census tract level 1970 Census to calculate neighborhood characteristics. I map census tracts to my 3 neighborhood categories (White, Mixed, and Black) based on the share of Black individuals in a given tract in 1970. I categorize a neighborhood as White if the Black share was less than 10%, Mixed if its Black share was between 10%-50%, and Black if its Black share was at least 50% in 1970.

I defined the consumption value of housing in neighborhoods ($\eta_{n,own}$ and $\eta_{n,rent}$) as the average number of rooms per neighborhood-tenure cell. I set the relocation cost k to \$150, which was the average rental cost of a moving truck for a day in 1970. To make the model estimation more tractable, as a normalization, I set the race-neutral amenity for households living in the White neighborhood (A_W) as -0.000250 . I normalize the three race-dependent neighborhood taste parameters of White agents ($\kappa_{W,W}$, $\kappa_{M,W}$, $\kappa_{B,W}$) to 1, and I assume that Black households gain 10% less utility from living in the White neighborhood than White families, which yields $\kappa_{B,W} = 1.1$. This assumption is in line with the findings of

Table 3: Externally estimated parameters

Parameter		Value	Source
Preferences			
Risk aversion	γ	2.0	Standard
Discount factor (non-hand-to-mouth)	$\beta_{HtM=0}$	0.96	Aguiar et. al. (2023)
Discount factor (hand-to-mouth)	$\beta_{HtM=1}$	0.72	Aguiar et. al. (2023)
Expenditure share of consumption	α	0.825	Brandsaas (2021)
Bequest parameter (simplified)	θ_2	100	
Share of White HtM agents		0.1214	SCF+
Utility from housing good: Owners	$\eta_{W,own}/\eta_{M,own}/\eta_{B,own}$	5.83/5.60/5.60	1970 Census
Utility from housing good: Renters	$\eta_{W,rent}/\eta_{M,rent}/\eta_{B,rent}$	4.33/4.18/4.14	1970 Census
Neighborhoods and housing market			
Housing depreciation	δ	0.025	Standard
House selling/buying cost	e_s/e_b	0.075/0.025	Brandsaas (2021)
Price elasticity of housing supply	Δ	1.37	Saiz (2010)
House price: White neigh.	p_W	\$22,198	1970 Census
House price: Mixed neigh.	p_M	\$16,339	1970 Census
House price: Black neigh.	p_B	\$13,567	1970 Census
Rent parameters: Intercept (W/M/B)		\$659/\$527/\$465	1970 Census
Rent parameters: Slope (W/M/B)		3.35%/2.85%/3.65%	1970 Census
Relocation cost	k	\$150	see text
Share with relocation barrier	π_F	0.5	see text
Mortgage and financial market			
Risk-free rate	i	0.02	Standard
Loan-to-value ratio	LTV	see text	SCF+
Debt-to-income ratio	DTI	0.1446	PSID
Mortgage length		24 years	1970 Census of Housing
Black mortgage discrimination	ψ_B	0.6329	Schafer & Ladd (1981)
75th percentile of household income (1970)	y_{75}	\$13,580	SCF+

the 1976 wave of the Detroit Area Study (Farley and Schuman, 1984) and Aliprantis et al. (2024), which found that, all else equal, Black households prefer to reside in Mixed and majority-Black neighborhoods over majority-White ones. I estimate the remaining two race-independent neighborhood amenities (A_M, A_B) and two race-dependent neighborhood taste parameters ($\kappa_{M,B}, \kappa_{B,B}$) within the model. Following Gordon (2023), I set π_F , the fraction of Black households living in the Black or Mixed neighborhood who are not allowed to move into the White neighborhood, to 50%.

Housing market I set neighborhood-level house prices p_n to the average of the median house prices in census tracts. Following Brandsaas (2021), I set the selling cost of a home

property e_s to 7.5% of the price, and the buying cost e_b to 2% of the price. House value depreciates with $\delta = 2.5\%$ between periods. The housing supply is modeled as

$$S_n = \bar{S}_n p_n^\Delta$$

where I set Δ to 1.37, the population-weighted mean of house supply elasticities for cities in my sample, originally calculated by [Saiz \(2010\)](#). S_n is the fraction of household living in neighborhood n , which is defined by the spatial equilibrium. I calculate the supply shifter parameter $\bar{S}_n$ to make sure that at spatial equilibrium, the equilibrium house prices p_n are equal to house prices computed from the 1970 Census. I calibrate rent in different neighborhoods by estimating the $rent_{i,n} = \beta_0 + \beta_1 p_{i,n} + \varepsilon_i$ equation in the three neighborhood types separately, using tract-level median home prices and rents from the 1970 census. Then, I set neighborhood-level rent q_n using the estimate parameters as $q_n = \hat{\beta}_0 + \hat{\beta}_1 p_n$, where p_n is the equilibrium house price of the model.

4.2 Parameters identified within the model

I jointly estimate the 8 remaining parameters of the model by targeting 8 moments with the method of simulated moments. I estimate the remaining two race-independent neighborhood amenities (A_M, A_B) and two race-dependent neighborhood taste parameters ($\kappa_{M,B}, \kappa_{B,B}$) within the model, and I identify them with four moments targeting the spatial distribution of Black households in White and Mixed, and White households in Mixed and Black neighborhoods. During estimation, I assumed that agents gain the highest utility by living in the White neighborhood and the lowest utility by living in the Black neighborhood of the model (thus, $A_W > A_M > A_B$), which reflects the existing differences in house prices and rents. Then, I identify the homeownership premium χ and the mortgage rate i^m parameters by targeting the Black and White homeownership rates. Finally, I target the final two

moments, the median of the Black and White wealth distributions, with the ratio of Black hand-to-mouth agents and with the White agents' bequest parameter $\theta_{1,W}$. I define Black agents' bequest parameter $\theta_{1,B}$ as one-tenth of $\theta_{1,W}$; as based on the SCF+, the median of Black wealth distribution at age 20-24 – when agents enter the model with inherited wealth – was zero.

I estimate the model with simulated annealing, which minimizes the weighted squared distance of the simulated and empirical moments:

$$\mathbf{d}' \mathbf{W} \mathbf{d}$$

where $\mathbf{d}$ is a vector containing the squared difference of the simulated and empirical moments, and $\mathbf{W}$ is a weighting matrix. The weighting matrix contains the inverse of the bootstrapped variance of the empirical moments on the diagonal and zeros elsewhere.

Table 4: Internally estimated parameters and model fit

Description	Parameter	Value	Moment	Model	Data
Spatial equilibrium					
Amenity of living in M neigh.	A_M	-32.1e-5	% B living in White neigh.	7.80%	7.29%
Amenity of living in B neigh.	A_B	-36.5e-5	% B living in Mixed neigh.	21.97%	21.53%
Neigh. taste param, B living in M neigh.	$\kappa_{B,M}$	0.8848	% W living in Mixed neigh.	9.19%	9.64%
Neigh. taste param, B living in B neigh.	$\kappa_{B,B}$	0.7940	% W living in Black neigh.	1.70%	2.08%
Mortgage market and homeownership					
Homeownership premium	χ	1.035	B homeownership rate	38.0%	38.3%
Mortgage rate	i^m	4.1%	W homeownership rate	71.9%	69.2%
Bequest & HtM					
Share of hand-to-mouth, B		45%	B median wealth	\$1,257	\$1,237
Bequest parameter, W	$\theta_{1,W}$	-0.00425	W median wealth	\$14,658	\$15,119

Table 4 presents the estimated parameter values and the difference between the simulated and empirical moments. The estimated race-dependent neighborhood taste parameters reflect that, compared to Black agents, Whites gain $1/0.8848 - 1 = 13.0\%$ less utility by living in the Mixed and 25.9% less utility by living in the Black neighborhood. The estimated

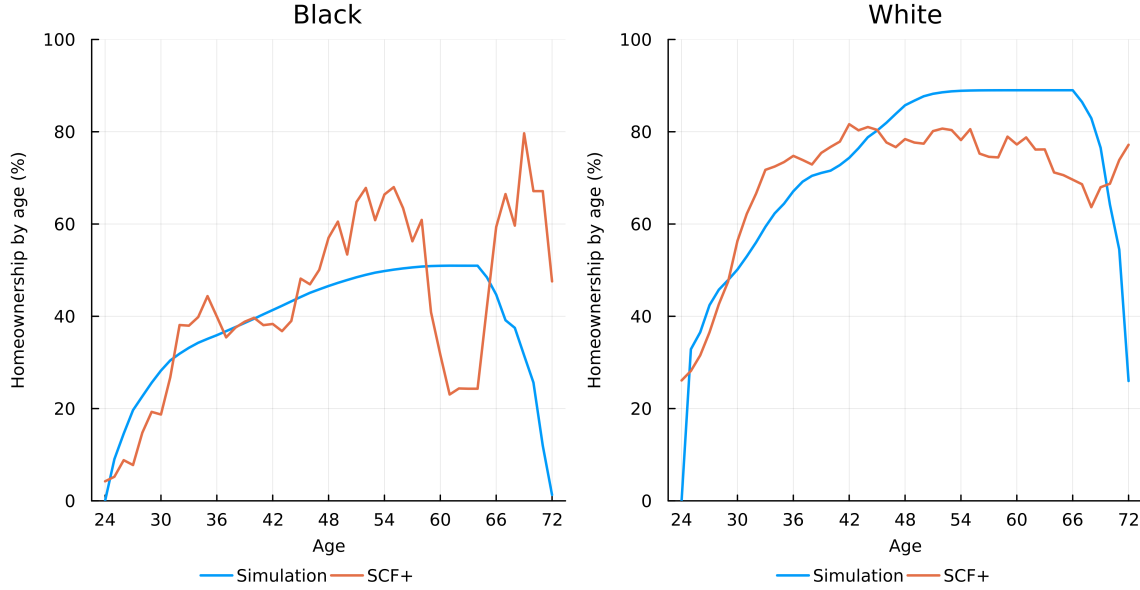
homeownership premium’s magnitude is in line with the literature (eg. 1.015 in [Kaplan et al. \(2020\)](#), and 1.379 in [Brandsaas \(2021\)](#)), while the mortgage premium ($i^m - i = 2.1\%$) is a little bit higher than in other papers (eg. 0.9% in [Brandsaas \(2021\)](#)). [Table 4](#) shows that the model matches most of the targeted moments quite closely. The simulated fractions of Black and White households living in different neighborhoods are within 0.5 percentage points of the targeted moments. The model matches the Black homeownership rate and median wealth moments well, while the median White wealth is nearly 3% lower than the targeted moment, despite that the generated White homeownership is 2.5 percentage points higher than in the data. Despite these differences, the simulated Black–White wealth gap at the median in the model is 11.7, which is close to the 12.2 ratio observed in SCF+.

To be able to run simulations, I set the $\bar{S}_n$ shift parameters in the model. Following [Chyn and Daruich \(2021\)](#), I use the estimated spatial distribution of households in the model to define the demand for housing, which is the fraction of households living in neighborhood n . The housing market clears in spatial equilibrium, when the housing supply S_n should be equal to the demand for housing, which allows me to calculate the supply shift parameters $\bar{S}_n$ using the spatial distribution of agents in the model and the 1970 house prices observed in the data. The calculated supply shift parameters guarantee that the model economy is in spatial equilibrium with the relocation barrier at the empirically observed house prices.

4.3 Model fit

The model intends to capture how spatial sorting and homeownership affect households’ wealth holdings. [Table 4](#) showed that the spatial distribution, homeownership rate, and median wealth holdings of both Black and White families in the estimated model fit well with their empirical counterparts. [Figure 5](#) and [Figure 6](#) illustrate that in the estimated model, the evolution of homeownership and median wealth over the life-cycle are also close to the observed life-cycle profiles in the SCF+.

Figure 5: Model fit: Homeownership over the life-cycle

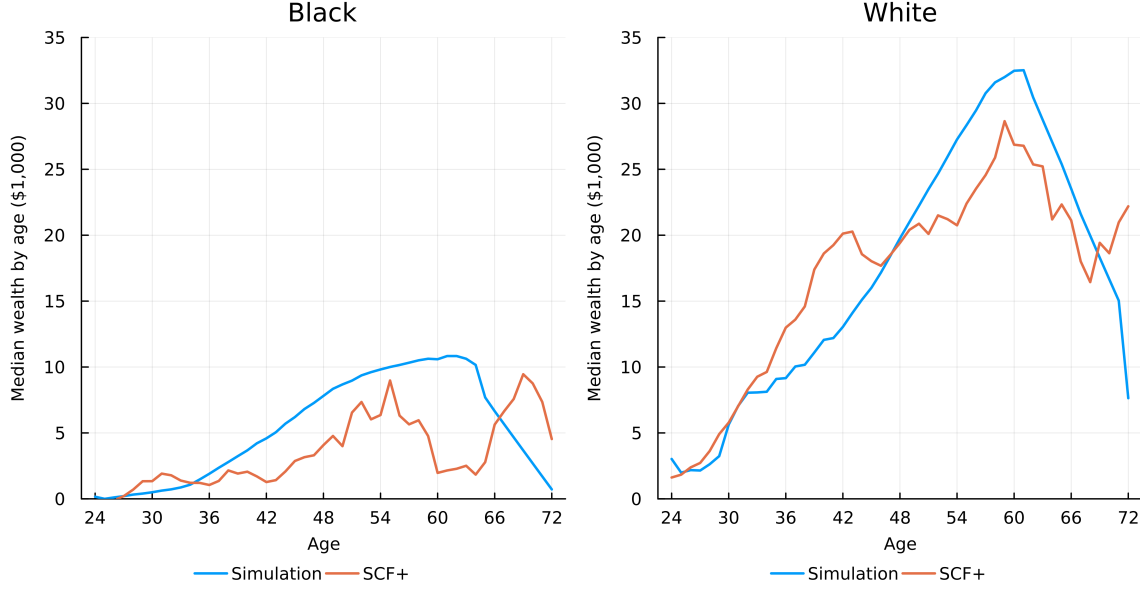


Source: own simulation and own calculation using SCF+. SCF+ data is moving averages (MA5) of homeownership rate at each age.

As Figure 5 shows, the model generates a realistic homeownership pattern for both Black and White agents. The model captures the fact that Black families' homeownership rate constantly increases over the life-cycle, while the drop in homeownership in the model after age 66 is the result of the certainty that agents leave the model at age 72. Meanwhile, the model captures the quick increase of the White homeownership rate at young age and the relatively flat homeownership rate between age 45-60.

As Figure 6 shows, the model generates slightly higher median Black wealth holding over the life-cycle than we can observe in the data. Meanwhile, the model generates a realistic wealth pattern over the life-cycle for White families. Compared to the empirical median wealth holding at the SCF+, the model slightly underpredicts wealth accumulation between ages 30-48 and overpredicts it between ages 48-60. However, the model captures the fact that the wealth holding of White families peaks around age 60, and then it starts to decline quickly.

Figure 6: Model fit: Median wealth over the life-cycle



Source: Own simulation, own calculation using SCF+. SCF+ data is moving averages (MA5) of median wealth at each age.

5 Simulations

I use this estimated model to simulate whether the Fair Housing Act could recreate the stylized facts I presented, and to quantify how the FHAct affected racial wealth inequality. I simulate the effect of the FHAct by lifting the relocation barrier in the model economy, which allows a π_F fraction of Black families living in Mixed or Black neighborhoods to move to the model's White neighborhood. Black families affected by this relocation friction could prefer to move into the White neighborhood for at least two reasons. First, the amenity level Black agents experience in the White neighborhood ($\kappa_{B,W}A_W$) is higher than in the Mixed ($\kappa_{B,M}A_M$) or in the Black ($\kappa_{B,B}A_B$) neighborhoods; otherwise, no Black agent would prefer to live in the White neighborhood, where the house price and rent are higher. Second, the consumption value of housing goods is also higher in the White neighborhood than in the Mixed and the Black, for both renters and owners (Table 3). As house prices and rents are

more expensive in the White neighborhood, Black agents need to forgo consuming consumption goods to pay the higher price of living there. Because of this forgone consumption, some Black agents with lower consumption levels might prefer to stay in the Mixed or the Black neighborhood, as the utility loss from moving and consuming fewer consumption goods could exceed the utility gain from higher amenity levels and higher housing service consumption. This is unlikely for high-income Black agents with higher consumption levels, for whom the net effect of moving on utility could be positive. Thus, the end of housing discrimination in the model does not affect all Black agents equally, and only higher-income Black agents are expected to move into the White neighborhood.

By lifting the relocation friction through the Fair Housing Act in the model, previously constrained higher-income Black agents start to move into the White neighborhood, which increases house prices and rents in neighborhoods that gain new residents and decreases house prices and rents in neighborhoods that lose population. Then, the changing house prices could motivate previously unconstrained Black and White families to move. In this way, the relocation of constrained Black families has a spillover effect on other families. At the end of this relocation process, the model reaches a new spatial equilibrium, at which no agent prefers to move at the equilibrium house prices. I quantify the effect of the Fair Housing Act on spatial sorting and racial wealth inequality by comparing the spatial and wealth distributions of the new steady state to those of the original state with relocation friction.

I compare the quantitative results of the simulations to the empirical changes in spatial distribution and housing wealth observed between 1970-80. I chose this comparison period for two main reasons: first, this is the shortest period following the Fair Housing Act between two decennial censuses, which is the data source of the spatial distribution of Blacks and Whites. Second, the model is estimated to match the economic conditions (such as income) observed in 1970, and choosing a longer comparison period than 1970-80 would have created

Table 5: Overview of the effect of the FHAct in the model economy

	Before FHAct	After FHAct	Change
Spatial distribution			
Fraction of Black agents living in			
White neighborhood	7.8%	13.8%	+6.0pp
Mixed neighborhood	22.0%	18.4%	−3.6pp
Black neighborhood	70.2%	67.8%	−2.4pp
Fraction of White agents living in			
White neighborhood	89.1%	88.3%	−0.8pp
Mixed neighborhood	9.2%	9.7%	+0.5pp
Black neighborhood	1.7%	2.0%	+0.3pp
Racial residential segregation			
Dissimilarity index (Black/White)	0.813	0.745	−0.068
Equilibrium house price in			
White neighborhood	\$22,198	\$22,214	+\$16
Mixed neighborhood	\$16,340	\$16,294	−\$44
Black neighborhood	\$13,567	\$13,531	−\$36
Average housing wealth			
Black	\$4,967	\$5,278	+\$311
White	\$12,576	\$12,642	+\$66
Homeownership			
Black	38.0%	38.1%	+0.1pp
White	71.9%	72.3%	+0.4pp
Average housing wealth of homeowners			
Black	\$13,073	\$13,864	+\$791
White	\$17,499	\$17,493	−\$6
Wealth inequality (Gini index)			
All families	47.44%	47.30%	−0.14pp
Black families	68.22%	68.42%	+0.20pp
White families	44.06%	43.91%	−0.15pp

a less realistic counterfactual comparison in the simulation results. The goal of the model is to quantify the effect of the Fair Housing Act by keeping all other relevant economic factors unchanged; thus, comparing the model’s simulation results to observed empirical changes becomes less and less realistic as we increase the time span.

As other important economic factors, aside from the Fair Housing Act, such as the real value of average total family incomes (Figure 4), also changed during the 1970s, the simulation results need not overlap with the observed changes in spatial distribution and housing wealth. The difference between the simulated and observed aggregates is informative

for determining whether the decreasing segregation and the rapidly increasing Black housing wealth are the consequence of the Fair Housing Act or of other economic changes in reality that are not incorporated directly into the model.

5.1 FHAct’s effect on spatial sorting

As [Table 5](#) shows, lifting the relocation barrier, which prevented some Black households from entering the White neighborhood in the model, increases Black presence in that neighborhood. In the new spatial equilibrium, the fraction of Black households living in the White neighborhood increases from 7.8% to 13.8%, primarily because previously constrained Black households move out from the Mixed and Black neighborhoods to the White neighborhood. As a result, the share of Black families living in the Mixed and Black neighborhoods decreases by 3.6 and 2.4 percentage points, respectively. Meanwhile, because of the relocation of Black families, house prices decrease in Mixed and Black neighborhoods, which motivates White families to move there. As a result, the fraction of White families living in the White neighborhood decreases by 0.8 percentage points, while the fractions living in the Mixed and Black neighborhoods increase by 0.5 and 0.3 percentage points, respectively. Because of the countermovement of White families into Mixed and Black neighborhoods, the fraction of agents living in each neighborhood remains relatively unchanged, and the change in equilibrium house prices is modest.

[Table 6](#) compares the change in spatial distribution of Black and White agents in the model to the observed change in two different distributions from the census. Panel B shows how the distribution of the Black and White population changed between 1970-80 in White, Mixed, and Black neighborhoods,⁴ which is a reliable comparison to the simulated changes in Panel A. The spatial distributions of renter and homeowner families are not available in

⁴Neighborhoods are defined based on Black share observed in 1970 for the 1970 census, and based on the Black share in 1980 for the 1980 census.

Table 6: Relocation across neighborhoods: Model vs. Data (1970-80)

	White neigh.	Mixed neigh.	Black neigh.
Panel A: Change in spatial distribution – Simulations			
Black agents	+6.0pp	-3.6pp	-2.4pp
<i>Homeowners</i>	+4.4pp	-2.0pp	-2.3pp
<i>Renters</i>	+1.6pp	-1.6pp	-0.1pp
White agents	-0.8pp	+0.5pp	+0.3pp
<i>Homeowners</i>	+0.1pp	+0.0pp	+0.3pp
<i>Renters</i>	-0.9pp	+0.5pp	+0.0pp
Panel B: Change in spatial distribution – Census, based on population change			
Black population	+3.0pp	+1.1pp	-4.2pp
White population	-2.3pp	+2.1pp	+0.1pp
Panel C: Change in spatial distribution – Census, based on occupied housing units			
Housing units occupied by Blacks	+3.0pp	+0.7pp	-3.7pp
<i>Homeowners</i>	+0.4pp	-0.7pp	+2.2pp
<i>Renters</i>	+2.6pp	+1.5pp	-6.0pp
Housing units occupied by Whites	-2.7pp	+3.1pp	-0.4pp
<i>Homeowners</i>	-1.6pp	+1.9pp	-0.1pp
<i>Renters</i>	-1.1pp	+1.2pp	-0.3pp

Notes: In census data, the White population contains both Hispanic and non-Hispanic Whites. Neighborhoods are defined based on the Black share observed in 1970 for the 1970 census and based on the Black share in 1980 for the 1980 census.

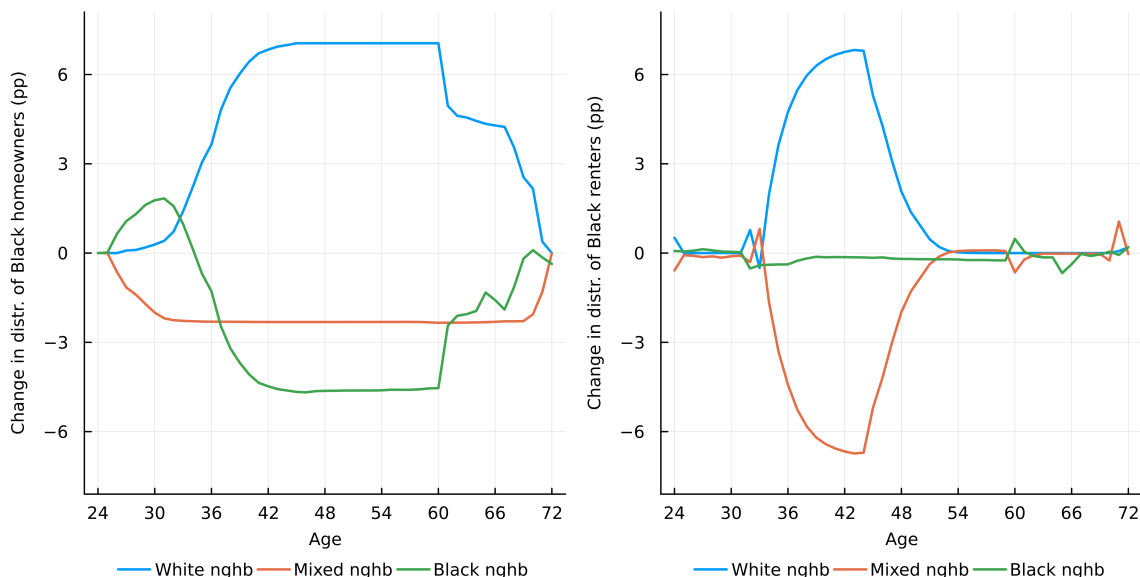
the census; only the distribution of housing units that were occupied by Black or White families. Panel C illustrates how the spatial distribution of housing units changed by race and tenure, which could be informative in determining whether homeowners or renters were more likely to move. In Panels A and C, agents and housing units are divided into renters and homeowners, and the sum of the percentage-point changes in homeowners and renters should add up to the total changes.⁵ Similarly, the total changes in the spatial distribution of Blacks and Whites should add up to zero among neighborhoods.⁶

As more than one family could have stayed in one housing unit, the two different mea-

⁵For example, Panel A shows that the share of Black families living in the White neighborhood increased by 6 percentage points. Of these 6 percentage points, the increasing share of Black homeowners living in White neighborhoods accounts for 4.4 percentage points, and the increasing presence of Black renters in White neighborhoods accounts for 1.6 percentage points.

⁶For example, in Panel A, the share of Black agents living in the White neighborhood increased by 6 percentage points and decreased by 3.6 percentage points in the Mixed and 2.4 percentage points in the Black neighborhood. These changes should add up to zero ($+6\text{pp} - 3.6\text{pp} - 2.4\text{pp} = 0\text{pp}$), or almost zero due to rounding errors.

Figure 7: Change of spatial distribution of Black agents over the life-cycle, by homeownership

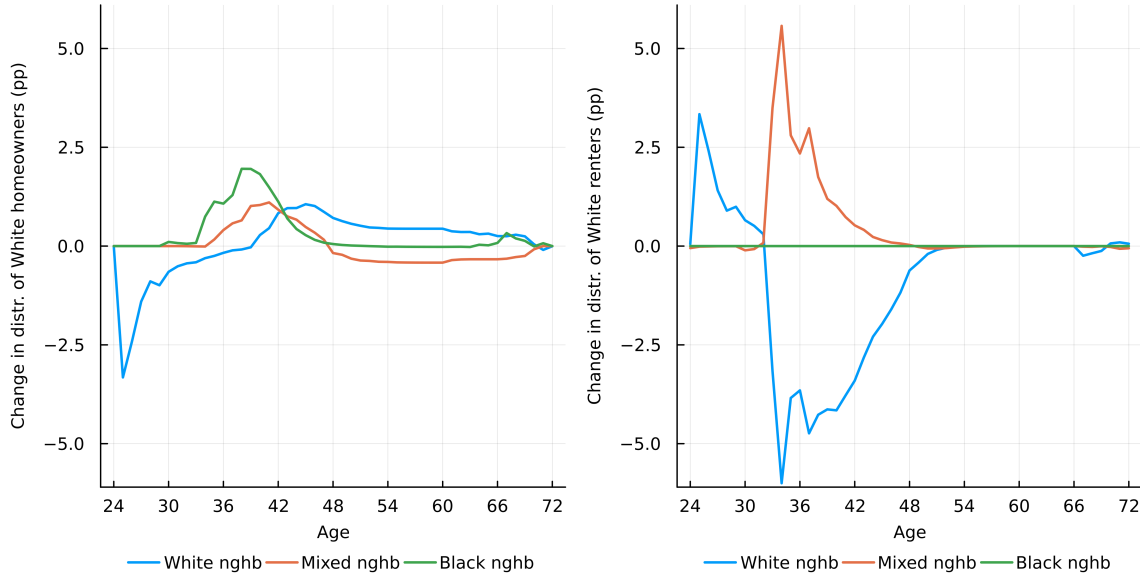


Source: Own simulation.

asures of spatial distribution change in Panel B and C in [Table 6](#) are quantitatively different, but they depict the same main patterns. Both measures suggest that between 1970 and 1980, the share of Black families living in White and Mixed neighborhoods increased, while the share of Whites living in Mixed neighborhoods increased by almost the same magnitude as the fraction of Whites living in White neighborhoods decreased. The only bigger difference between Panels B and C is the change in the share of the White population and housing units residing in Black neighborhoods. Even if the share of housing units occupied by White households decreased by 0.4 percentage points in Black neighborhoods (in Panel C), the share of Whites living in these neighborhoods increased by 0.1 percentage point (in Panel B).

Comparing simulation results in Panel A with the observed changes in the spatial distribution of Blacks and Whites in Panel B reveals that the Fair Housing Act could explain the direction of changes in the spatial sorting of Blacks and Whites. Simulation results

Figure 8: Change of spatial distribution of White agents over the life-cycle, by homeownership



Source: Own simulation.

suggest that, because of the Fair Housing Act, Black families relocated from Black to White neighborhoods, while the FHAct also explains why the share of Whites living in White neighborhoods decreased and their share in Mixed and Black neighborhoods increased. Compared with the observed change in spatial distribution, the model predicts lower White relocation from White to Mixed neighborhoods and higher Black relocation to White neighborhoods. Also, in the model, Black presence decreased in Mixed neighborhoods because of the Fair Housing Act, while in reality, the share of Blacks living in Mixed neighborhoods increased between 1970 and 1980. The source of these differences could be the unmodeled economic changes during the 1970s, or the possibility that neighborhoods could have become Mixed from White as Black families moved there, which the model cannot capture because of the limited number of neighborhoods.

The Fair Housing Act had a sizable effect on housing wealth through Black and White

agents' location and homeownership decisions in the model. [Figure 7](#) and [Figure 8](#) illustrate how FHAct changed the spatial distribution and homeownership decision of agents over the life-cycle. The right panel of [Figure 7](#) shows that the FHAct mainly affected Black renters' spatial distribution in the middle of agents' lives, as previously constrained Black agents moved into the White neighborhood from the model's Mixed neighborhood. Because of the unconstrained relocation following the FHAct, many previously constrained Black homeowners moved from the Mixed and Black neighborhood into the White neighborhood, but following retirement at age 60, many of these Black homeowners moved back to the Black neighborhood and consumed some part of the previously accumulated wealth.

Meanwhile, as [Figure 8](#) shows, the spillover effect of Black agents' relocation had a sizable effect on White agents' location and homeownership decisions in the first part of their lives. Following the FHAct, house prices and rents increased in the White and decreased in the Mixed and Black neighborhoods of the model. This motivated some White families to postpone homeownership in the White neighborhood at the beginning of their lives and become homeowners in the Mixed and Black neighborhoods. In the end, as the decreased share of White homeowners in the Mixed and their increased share in the White neighborhood suggest, some White homeowners were able to replace their home in the Mixed neighborhood with a home in the White neighborhood in the second part of their lives. These agents were able to accumulate more housing wealth than before the FHAct, which increased both the average housing wealth and the homeownership rate of White agents ([Table 5](#)).

5.2 Could FHAct have caused the three stylized facts?

To determine whether the Fair Housing Act caused the three stylized facts I presented in Section 2, I compare the simulated changes in spatial distribution and housing wealth in the model to the observed empirical changes between 1970-80. Choosing this short time window right after the FHAct yields realistic but not perfect comparisons, as, contrary to the model

economy, economic conditions changed in the real world.

Stylized Fact I: Decline of residential segregation

In line with the sizable changes in the spatial distribution of Black agents and the countermovement of Whites, the dissimilarity index between Black and White families decreased from 81.3% to 74.5% in the model (Table 7). The 6.8 percentage-point decline in the dissimilarity index is slightly higher than the 6.1 percentage-point decline observed in the data between 1970-80. This suggests that, by easing the relocation barriers Black families faced while moving to White neighborhoods, the Fair Housing Act could explain the start of residential segregation's decline during the 1970s.

Stylized Fact II: Black families started to leave Black neighborhoods

As Table 7 summarizes, the Fair Housing Act decreased the share of Black agents residing in Black neighborhoods in the model. The observed decline in the share of Blacks residing in Black neighborhoods (4.1 percentage points) was larger than the model prediction (2.4 percentage points), suggesting that factors not directly incorporated into the simulations prevented Black families from moving to White neighborhoods. At the same time, the simulation results verify that the Fair Housing Act increased the outmigration of Black families from Black neighborhoods.

Stylized Fact III: Black families' housing wealth increased at a faster pace than Whites'

Figure 1 in Section 2 shows that even if the nominal differences in Black and White average housing wealth increased during the 1970s (left panel), the growth rate of Black housing wealth was higher than that of White (right panel), which is the first necessary condition to close the racial gap in nominal housing wealth differences in the long run. As

Table 7: Stylized facts: Model vs. Data

	Data (1970-80)	Simulation
Stylized fact I: Decline of racial segregation		
Dissimilarity index (Black/non-Black)	-6.1pp	-6.8pp
Stylized fact II: Black families started to leave Black neighborhoods		
Change in the share of Black families living in Black neighborhoods	-4.1pp	-2.4pp
Stylized fact III: Black housing wealth increased at a faster pace than White		
Racial difference in avg. housing wealth growth (Black – White)	+30.2pp	+5.7pp
Change of racial gap in homeownership (Black – White)	+5.6pp	-0.3pp
Racial difference in avg. housing wealth growth of homeowners (Black – White)	+5.1pp	+6.3pp

Table 7 summarizes, the difference between the growth rate of Black and White average housing wealth between 1965-69 and 1975-79 (right panel of Figure 1) was 30.2 percentage points. In the model, the FHAct increased the average Black and White housing wealth by 6.26 and 0.52 percentage points, respectively, yielding a 5.7 percentage-point difference in racial housing wealth growth. Thus, the simulation results suggest that the Fair Housing Act increased Black housing wealth at a faster pace than that of Whites, and that the magnitude of the FHAct’s direct effect is almost a fifth of the observed racial difference in the housing wealth growth rate.

We can divide the change in average housing wealth into the change in homeownership rate and the change in average housing wealth of homeowners. This distinction reveals that in the model, Black housing wealth increased faster than White because Black homeowners increased their housing wealth more than White homeowners. Following the FHAct in the model, many Black families decided to own homes in White rather than Mixed or Black neighborhoods (see Figure 7), thereby increasing their housing wealth. Meanwhile, the average housing value of White homeowners slightly decreased in the model, primarily because 0.3% of White agents switched from being renters in the White neighborhood to becoming homeowners in the cheaper Black neighborhood (Table 6). As a result, in the model, the growth rate of Black homeowners’ housing wealth was 6.3 percentage points higher than that

of White homeowners', which is quite similar to the observed difference in the data (+5.1 percentage points).

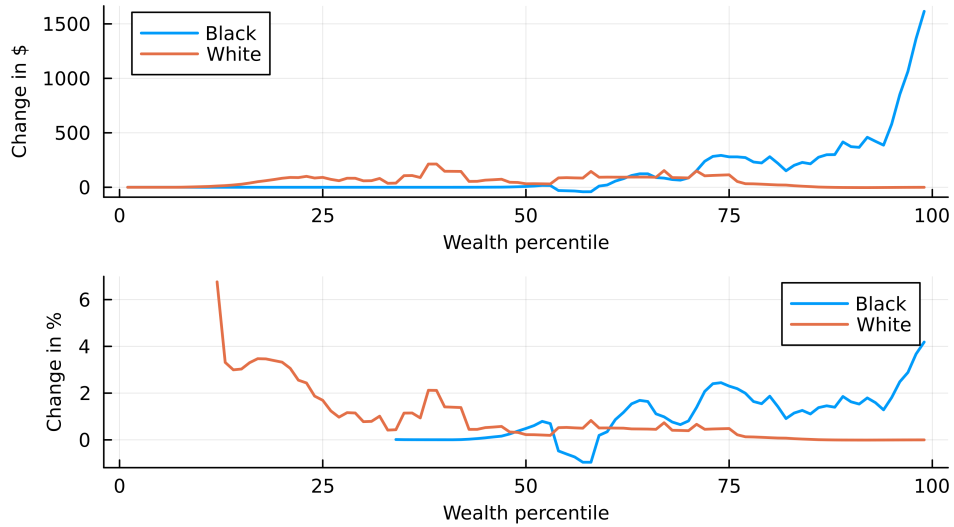
In the model, the Fair Housing Act increased the White homeownership rate (+0.4pp) at a faster pace than the Black homeownership rate (+0.1pp); as a result, racial differences in homeownership rates increased by 0.3 percentage points due to the FHAct. In reality, the Black homeownership rate increased faster than the White rate (see [Figure 2](#)), and racial differences in the homeownership rate decreased by 5.6 percentage points. The simulation results suggest that the observed large decline in the racial homeownership gap was not the direct effect of the Fair Housing Act but rather the result of other favorable changes in economic conditions (such as rising real incomes).

All in all, my simulation results suggest that the Fair Housing Act decreased racial differences in average housing wealth, but not at the magnitude observed in the SCF+. However, because the narrowing gap resulted from relatively wealthy Black homeowners relocating from the Mixed and Black neighborhoods to the White one, the declining racial differences in housing wealth might not imply a reduction in racial wealth inequality across most of the wealth distribution.

5.3 FHAct's effect on racial wealth inequality

Finally, I quantify the effect of the Fair Housing Act on racial wealth inequality by comparing changes in the steady-state of Black and White wealth distributions. [Figure 9](#) illustrates how the wealth holding changes at a given percentile of the Black and White wealth distribution, both in dollar and percentage terms. Based on my simulation results, the Fair Housing Act had a sizable and heterogeneous effect on the Black wealth distribution. Wealth holding in the top 25% of the Black wealth distribution increased by 1.5–4%, while in the 6th decile, wealth holding decreased by almost 1%. These results suggest that the positive effect of the FHAct was concentrated at the upper end of the Black wealth distribution, while the

Figure 9: Effect of free movement on wealth percentiles

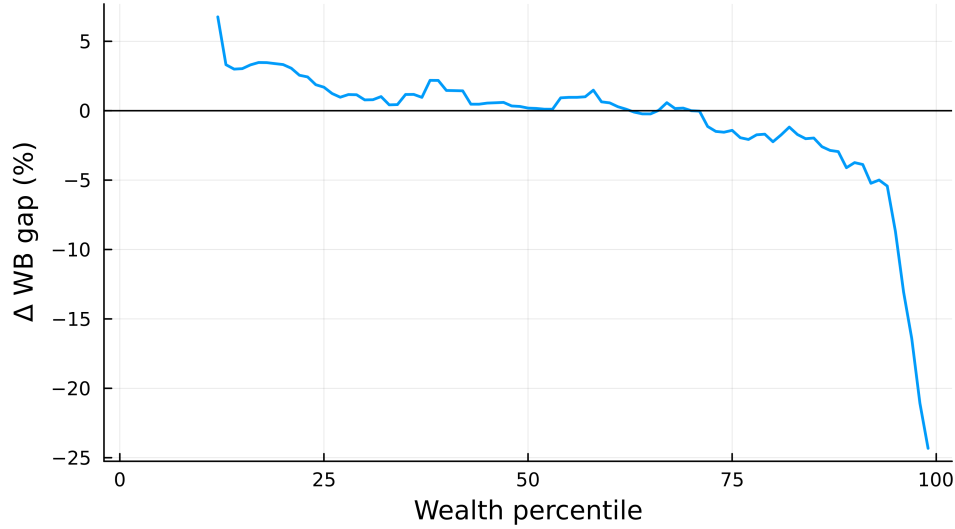


Source: Own simulation. Lines are moving averages (MA5) of the change of simulated wealth percentiles, defined as $p' - p$, where p' is a given wealth percentile without the relocation barrier, and p is the same wealth percentile with the barrier. Missing data points at the bottom were caused by division by zero wealth.

FHAct could have had a negative effect on the wealth holdings of families in the middle of the Black wealth distribution. As a result, the GINI index among Black agents increases from 68.2% to 68.4% in the model, and thus the FHAct slightly increased wealth inequality among Black families.

Meanwhile, the Fair Housing Act increased wealth holdings at the bottom and middle of the White wealth distribution. The top panel of Figure 9 shows that even if the change in White wealth holding did not exceed \$200 in any wealth percentiles, wealth holding increased by 2–6% around the 15–40th percentiles of the White wealth distribution. One source of this change is the increasing White homeownership in Black neighborhoods at relatively early ages (Figure 8), as some White agents switched from being renters in the White to being homeowners in the Black neighborhood. Also, as Figure 8 shows, this switch allowed a small fraction of White agents to own a home in the White instead of the Mixed neighborhood at later ages. So, even if White agents were not directly affected by the FHAct in the model, the policy's implied house price changes alter the optimal homeownership and

Figure 10: Effect of free movement on White/Black wealth gap at different percentiles



Source: Own simulation. Lines are moving averages (MA5) of the percent change of simulated White/Black wealth gap at different percentiles; see text for further clarification of calculation. Missing data points at the bottom were caused by division by zero wealth.

location choices of White agents at the margin of becoming homeowners. As these agents are concentrated in the middle of the White wealth distribution, the Fair Housing Act's positive effects are concentrated between the 20th and 60th percentiles. As the top of the White wealth distribution remained unaffected, wealth inequality declined slightly among White families (Table 5).

We can quantify the Fair Housing Act's effect on racial wealth inequality by comparing how the nominal differences between the Black and White wealth distribution changed at each wealth percentile:

$$\Delta WB gap_p = \frac{WB gap_{p,post}}{WB gap_{p,pre}} = \frac{W_{p,post} - B_{p,post}}{W_{p,pre} - B_{p,pre}}$$

where W_p and B_p are the wealth holding of the White and Black wealth distribution at percentile p , WB_p is the nominal difference between the White and Black wealth distribution at percentile p , while *pre* and *post* reflect the equilibrium before and after the introduction

of the FHAct in the model. Increasing $\Delta WB gap_p$ means that the Fair Housing Act increased nominal wealth differences between White and Black households at percentiles p . As [Figure 10](#) depicts, the Fair Housing Act decreased racial wealth inequality at the top quarter of the wealth distribution, especially in the very top percentiles, where the decrease reached double digits. As the FHAct did not affect the top of the White wealth distribution in the simulation, the large reduction of racial wealth inequality at the top is the result of relocating Black families, who accumulated more net housing wealth than before. Meanwhile, the Fair Housing Act slightly increased racial wealth inequality between the 2nd and 7th wealth deciles, where $\Delta WB gap_p$ increased by 1–2%, on average.

As [Figure 9](#) shows, the FHAct decreased wealth holdings in only a small part of the Black wealth distribution; for the majority of households, the policy change was wealth-neutral or increased their wealth. In this sense, the FHAct increased the wealth and the utility of almost all agents in the model. However, my simulations reveal that the gains from the FHAct were quite unequal, with most of them concentrated at the top of the Black wealth distribution and in the middle of the White wealth distribution. As a result, the FHAct increased wealth inequality among Black families and decreased it among White families. Due to the policy change, wealthy Black families were able to get closer to wealthy White families' wealth holdings, but the FHAct also increased the wealth gap between Black and White households across most of the wealth distribution. Put another way, based on my simulation results, even if the vast majority of Black agents experienced increasing or unchanged wealth due to the FHAct, more than half of Black agents experienced that White agents at the same percentile of their wealth distribution benefited more from the policy change than they did.

6 Conclusion

In 1968, the Fair Housing Act (FHAct) terminated lawful housing discrimination, which was widely regarded as one of the major sources of racial wealth inequality. In this paper, I investigate whether, in the absence of other changes in the economy, the Fair Housing Act reduced racial residential segregation and racial wealth inequality simultaneously. First, I show empirically that following the FHAct, more and more Black families moved to integrated neighborhoods, and racial residential segregation declined quickly. At the same time, despite Black families accumulated housing wealth at an accelerated pace during the 1970s, wealth inequality between Black and White families did not change markedly in the following decades ([Derenoncourt et al., 2024](#)).

I quantify the direct effect of the FHAct on segregation and racial wealth inequality by estimating a quantitative heterogeneous agent life-cycle OLG model. Among the first in the literature, I jointly model households' homeownership and location choices in a dynamic framework, yielding new insights into whether the FHAct motivated households to change neighborhoods to become homeowners at different points in their lifetimes. In the model, households' relocation among neighborhoods endogenously changes house prices and, through them, housing wealth, while changes in house prices create a spillover by motivating other households to move. I model housing discrimination by preventing a fraction of Black families from moving into White neighborhoods, while White households are free to move into any neighborhood of the model. I estimate this model to match the observed spatial distribution, homeownership, and median wealth holding of Black and White families in 1970.

I simulate the effect of the FHAct by allowing all Black families to move into the model's White neighborhood, which motivated 6% of Black households to move there. The relocation of predominantly high-income Black households increased house prices in the

White neighborhood and decreased house prices in the Mixed and Black neighborhoods, which motivated 0.8% of White agents to leave the White neighborhood of the model. As a result, I find that the FHAct reduced the most common measure of segregation, the dissimilarity index between Blacks and non-Blacks, by 6.8 percentage points, which is slightly higher than the observed decline in segregation during the 1970s. Meanwhile, the FHAct also reduced racial differences in average housing wealth but had a heterogeneous effect on racial wealth inequality. I find that most wealth gains under the Fair Housing Act were concentrated in the upper part of the Black wealth distribution and the middle part of the White wealth distribution. Therefore, racial wealth inequality decreased substantially in the top 25% but increased slightly in the middle of the wealth distribution.

My results suggest that, in general, the Fair Housing Act did not lead to a decline in racial wealth inequality; moreover, I find that racial wealth inequality increased in the majority of the wealth distribution due to the Fair Housing Act. As an important policy implication, my work illustrates the need to account for heterogeneous general-equilibrium effects when designing new policies aimed at creating a more equal society by changing the spatial distribution of families. As my findings suggest, most of the positive effects of these policies could easily be concentrated among already wealthy, non-targeted families as an unintended consequence.

For tractability, I made several simplifying modeling assumptions. One of these simplifications is the limited and unchanging number of neighborhoods, which prevents White agents from relocating to another White neighborhood as Black families move into the model's only White neighborhood following the Fair Housing Act. Another simplification is the exogeneity of neighborhood amenities, implying that amenities do not change as racial composition or average income changes in neighborhoods. Easing these simplifications in future work could yield a more complex picture of the connections among housing discrimination, residential segregation, and racial wealth inequality.

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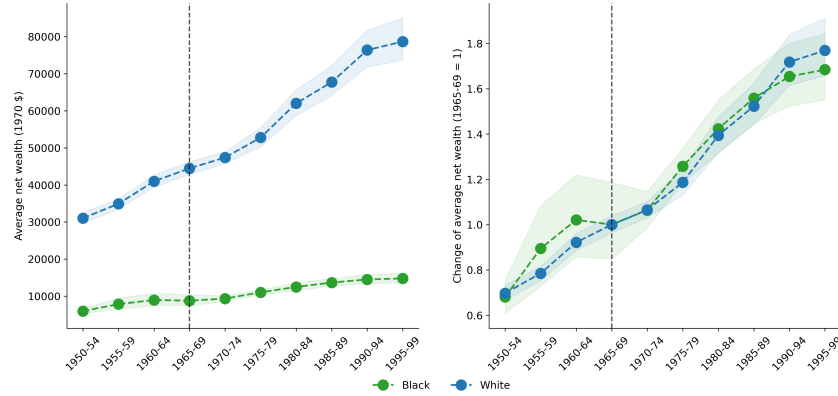
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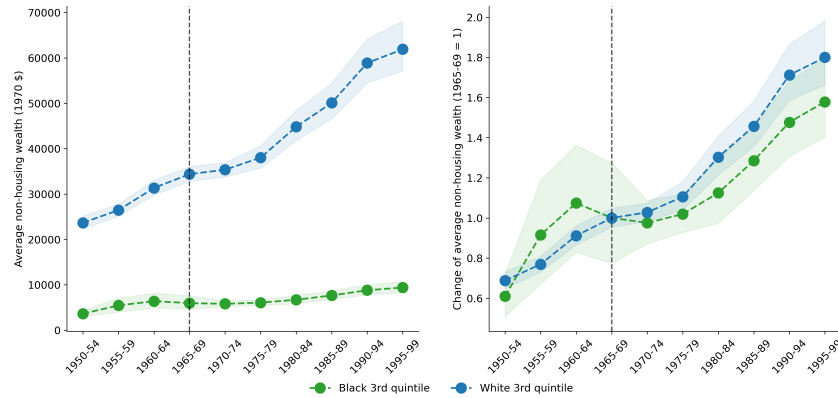
Appendix A: Stylized Facts

Figure A1: Evolution of average net wealth of the Black and White wealth distribution



Source: own calculation, based on SCF+ data. Net wealth is defined as the difference between total asset value minus existing debt. Prices measured in 1970 dollars. Transparent areas represent bootstrapped 95% confidence intervals.

Figure A2: Evolution of average non-housing wealth of the Black and White wealth distribution



Source: own calculation, based on SCF+ data. Non-housing wealth is the difference between total net wealth and housing wealth, which is defined as asset value minus the remaining mortgage of the first residence. Prices measured in 1970 dollars. Transparent areas represent bootstrapped 95% confidence intervals.

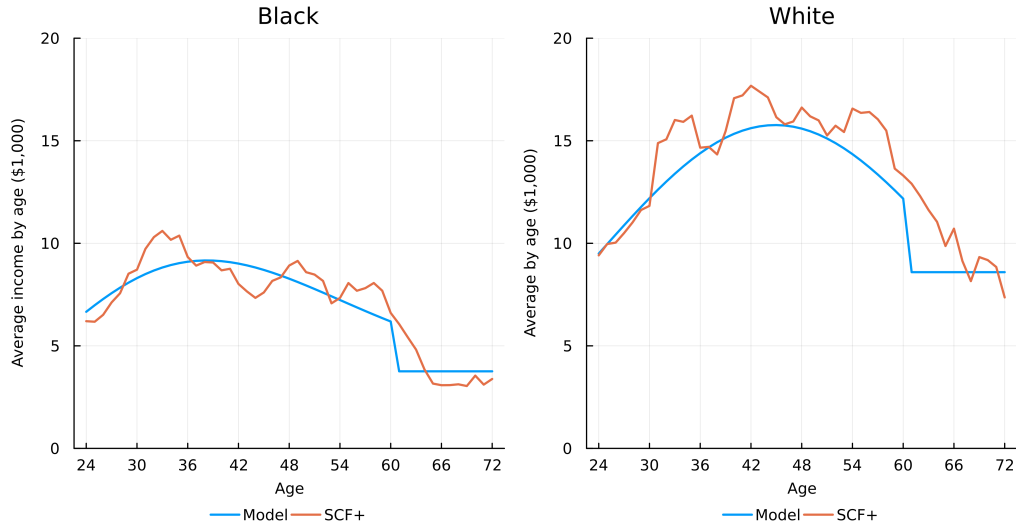
Appendix B: Estimation

Table A1: Calibrated parameters of the income process

	Black income distribution	White income distribution
Life-cycle elements		
Age^0	5.879769	7.155332
Age^1	.1877262	.1076763
Age^2	-.0036298	-.0012546
Age^3	.0000205	8.21e-07
Income fixed effect ρ_R		
Low	-.8481999	-.6473573
Medium	.0899008	.0929006
High	.7667469	.5564716
Temporary income shock parameter with Low ρ_R: $\epsilon_{\rho=L,R}$		
Low	-.3672556	-.2847186
Medium	.0019360	.00606490
High	.3653196	.2787603
Temporary income shock parameter with Medium ρ_R: $\epsilon_{\rho=M,R}$		
Low	-.2831905	-.2100579
Medium	.0118372	-.0003271
High	.2717945	.2105470
Temporary income shock parameter with High ρ_R: $\epsilon_{\rho=H,R}$		
Low	-.2307297	-.2016939
Medium	.0115538	.00829980
High	.2215885	.1935596

Notes: Own calculation based on PSID and SCF+ data. See text for more details.

Figure A3: Fit of calibrated income distribution



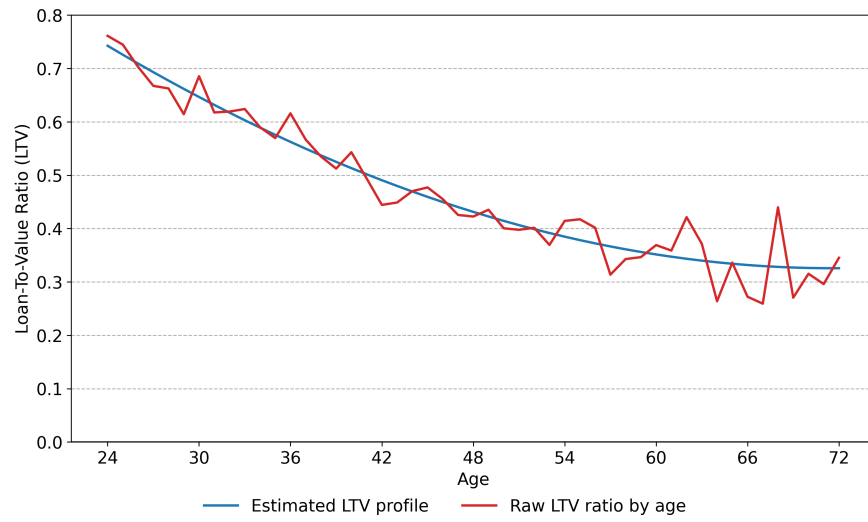
Source: own calculation, based on SCF+ and PSID data. Income is defined as total family income. Prices measured in 1970 dollars.

Table A2: Marginal tax rates in 1970 for married couples filling jointly

Over	But not over	Marginal tax rate	Over	But not over	Marginal tax rate
0	1000	14%	40,000	44,000	48%
1,000	2,000	15%	44,000	52,000	50%
2,000	3,000	16%	52,000	64,000	53%
3,000	4,000	17%	64,000	76,000	55%
4,000	8,000	19%	76,000	88,000	58%
8,000	12,000	22%	88,000	100,000	60%
12,000	16,000	25%	100,000	120,000	62%
16,000	20,000	28%	120,000	140,000	64%
20,000	24,000	32%	140,000	160,000	66%
24,000	28,000	36%	160,000	180,000	68%
28,000	32,000	39%	180,000	200,000	69%
32,000	36,000	42%	200,000		70%
36,000	40,000	45%			

Source: 1970 Instructions for Form 1040, Schedule Y, <https://www.irs.gov/pub/irs-prior/i1040--1970.pdf>, accessed at 10/24/2024.

Figure A4: Estimated LTV ratio



Source: own calculation, based on SCF+. Estimated cubic function: $LTV = (123.8819 - 2.425384 \times age + 0.0142931 \times age^2 + 0.0000247 \times age^3)/100$.